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Fuzzy harmonic mean technique for solving fully fuzzy multilevel multiobjective linear programming problems

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Abstract This paper illustrates how the fuzzy harmonic mean technique can efficiently solve fully fuzzy multilevel multiobjective linear programming (FFMMLP) problems. First, at each level, the FFMMLP problem can be converted into three crisp multiobjective linear programming (CMLP) problems using the crisp linear technique. Then, the fuzzy harmonic mean technique is utilized to aggregate each crisp problem's multiobjective into a single objective. Second, the ensuing final, single-objective problem is constructed using the harmonic mean for each level. Finally, it is solved to obtain a fuzzy compromise solution for the FFMMLP problem in general. Two examples are given to obtain the application of the proposed method. One example is applying the proposed approach to a multilevel multiobjective production planning model for a supply chain under a fully fuzzy environment.

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1. Introduction

The concept of fuzzy decision-making was brought into programming research by Zimmermann [36] under the term fuzzy linear programming (FLP), and it has since been used in a variety of problem types [16,30]. Abou-El-Enien and El-Feky [2] extended the TOPSIS approach to finding preferred solutions for multilevel multiobjective linear programming problems

under a fuzzy environment. Pramy [26] used the graded mean integration representation method to convert the fuzzy multiobjective linear fractional programming problem into a crisp multiobjective linear fractional programming problem. The proposal given by Osman et al. [24] is used to solve the multilevel multiobjective fractional programming problem with fuzzy demands. If all parameters and decision variables in an FLP problem are allowed to be fuzzy, the problem's model can be remodeled to fully fuzzy linear programming (FFLP) [7,8]. Based on the bound and decomposition method and the separable programming method, Abdalhakim et al. [1] presented an approach to solving a fully fuzzy multilevel quadratically constrained quadratic programming problem. Emam et al. [9] proposed an interactive approach based on the

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bound-decomposition technique and ϵ -constraint method to find the optimal fuzzy solution of the fully fuzzy multilevel linear programming problem. Bharati et al. [5] introduced a class of distance functions between two trapezoidal fuzzy numbers and used it to solve a fully fuzzy multiple objective linear programming (FFMOLP) problem. A fuzzy preferred solution for FFMOLP problems was developed by Hamadameen & Hassan [11]. Arana-Jimenez [4] introduced a method based on multiobjective linear programming to find an optimal solution to FFLP problems.

Fuzziness can be represented in various forms. One of the most suitable forms is a membership function. Additionally, based on the description or type of membership function, a fuzzy number can be categorized in various ways, such as a triangular fuzzy number (TFN) or trapezoidal fuzzy number. TFNs are often used in applications [22,28]. Zhang et al. [34] proposed a new method to measure the similarity of two TFNs and applied the TFN in a collaborative filtering recommendation system. Voskoglou [31] used TFNs as a tool for student assessment. Wan [32] studied the type recognition problem under a triangular fuzzy number environment and applied it for multisensor object recognition.

Multilevel programming (MLP) techniques are formulated to solve decentralized problems with multiple decision-makers organized at hierarchical levels [17]. Several mathematical formulations for such problems have been presented [10,13]. In their work, Han et al. [12], bilevel and trilevel programming problems were solved using particle swarm optimization. The authors also introduced a survey of the approaches and the theoretical results to MLP [19]. Osman et al. [23] developed an interactive approach to find a preferred solution for optimizing the multilevel multiobjective fractional programming problem under hybrid uncertainty. Zhang et al. [35] developed an algorithm to solve a decentralized tri-level decision-making problem and applied the proposed algorithm in the supply chain's third-party logistics decision-making.

Since the publication of the early work of Chandra [6] on the concept of multiple-objective programming (MOP) problems, many researchers have proposed methods to solve MOP problems and find Pareto solutions for them [3,20]. Sulaiman and Mustafa [29] suggested an algorithm based on a harmonic mean technique for solving multiobjective linear programming (MOLP) problems. Muruganandam and Ambika [21] introduced a new method for multiple objective linear fractional programming problems based on a graded mean integration representation and a harmonic mean technique. Sohag and Asadujjaman [27] suggested a new average method algorithm to solve MOLP problems. Pei [25] proposed transformation characteristics to address multiobjective linear fractional programming problems.

To the best of our knowledge, no method based on the harmonic mean technique thus far has been presented with FFMMLP problems. This leading research gap leads to two questions that need to be answered:

- Is there any method that considers FFMMLP problems?
- If there is such a method, can we use it to solve complicated applications?

Searching for the preferred solution for FFMMLP problems is a complicated task. Various approaches have been developed to solve multilevel multiobjective linear program-

ing problems in a fully fuzzy environment without the harmonic mean technique. Each of these approaches is Fathy [10]. He proposed an interesting approach based on the fuzzy approach to fully fuzzy multilevel multiobjective integer quadratic programming problems. Our article's proposed methodology is only one of the procedures based on the harmonic mean technique introduced for solving FFMMLP problems.

This work can be applied to real-life applications such that supply chain management, network interdiction, traffic and transportation network design, safety and accident management, policy making, energy management, taxation problems, capacitated lot-sizing problems, and so on. For example, it may be applied on solving the airport slot-scheduling problem, since the growth rate of the demand for transport services is expected to exceed the growth rate of airport infrastructure development and the airport's capacity [37]. Another real life application is sustainable water system planning and Arid agricultural regions [38]. Since, all over the world, the water scarcity crisis threatens food production and sustainable development, especially for arid agricultural areas. Therefore, it is necessary to develop plans and strategies to manage agricultural water in a sustainable manner and improve the efficiency of water management, as there are many complications in agricultural water management in arid areas that affect decision-making levels. Multiple targets, water users and uncertainty.

This article's work is based on the concepts discussed in [29]. In [29], the authors examined the harmonic mean technique to deal with multiobjective optimization. However, the existing method [29] cannot be applied to obtain the preferred solution of the FFMMLP problems. In this work, we develop the same concept by explaining new methodological modifications in fully fuzzy data. The modified harmonic mean technique will be expected to be applied to real-life applications in various areas.

The main contributions of our presented method are as follows:

- A new approach is presented to address fully fuzzy multilevel multiobjective linear programming (FFMMLP) problems. All decision variables and parameters are characterized by triangular fuzzy numbers.
- Utilizing a crisp linear technique [14], the presented problems are transformed into three crisp multilevel multiobjective linear programming (MMLP) problems.
- Utilizing the fuzzy harmonic mean technique [33], the crisp problem's multiple objectives are aggregated into a single objective.
- Using the harmonic mean technique [29] and standard simplex algorithms, a single-objective problem is constructed and then solved to obtain a fuzzy compromise solution for FFMMLP problems.
- The proposed approach is applied to the multilevel multiobjective production planning model for a supply chain under a fully fuzzy environment.

In addition, we shall mention that the proposed method cannot be applied to find a compromise solution of fully rough multilevel multiobjective linear programming problems and fully rough multilevel multiobjective linear fractional programming problems. Therefore, this method may be extended to overcome these shortcomings in future research.

This article is constructed into six sections. In [Section 2](#), definitions of triangular fuzzy numbers and the harmonic mean technique are necessary for the discussion. In [Section 3](#), FFMMLP problems are modeled, and a detailed methodology is given to solve this situation. [Section 4](#) introduces the steps of solving the proposed problems, and [Section 5](#) provides numerical examples of the proposed method. Finally, conclusions are drawn in [Section 6](#).

2. Preliminaries

2.1. Fuzzy numbers

This section gives necessary notations related to triangular fuzzy numbers (TFNs) used in this work.

Definition 1. [15] Let $\tilde{b} = (b_1, b_2, b_3)$ and $\tilde{c} = (c_1, c_2, c_3)$ be two TFNs. The basic arithmetic operations are given as follows:

- Subtraction: $\tilde{b} - \tilde{c} = (b_1 - c_3, b_2 - c_2, b_3 - c_1)$.
- Addition: $\tilde{b} + \tilde{c} = (b_1 + c_1, b_2 + c_2, b_3 + c_3)$.
- Scalar multiplication: $k\tilde{b} = (kb_1, kb_2, kb_3)$ if $k \geq 0$,
 $k\tilde{b} = (kb_3, kb_2, kb_1)$ if $k < 0$.
- Multiplication:

$$\begin{aligned} \tilde{b} \times \tilde{c} &= (b_1c_1, b_2c_2, b_3c_3), & b_1 \geq 0, \\ \tilde{b} \times \tilde{c} &= (b_1c_3, b_2c_2, b_3c_3), & b_1 < 0, \quad b_3 \geq 0, \\ \tilde{b} \times \tilde{c} &= (b_1c_3, b_2c_2, b_3c_1), & b_3 < 0. \end{aligned}$$

Definition 2. [15] Let $\tilde{b} = (b_1, b_2, b_3)$ and $\tilde{c} = (c_1, c_2, c_3)$ be two TFNs; then, the partial order relation $\succ$ can be defined as follows:

$\tilde{b} \succ \tilde{c}$ iff $b_i \geq c_i, i = 1, 2, 3$ and $b_k > c_k$ for some $k \in \{1, 2, 3\}$.

2.2. Fuzzy harmonic mean

Definition 3. [33] Let $WHM : (R^+)^n \rightarrow R^+$, if

$$WHM(b_1, b_2, \dots, b_n) = \frac{1}{\sum_{j=1}^n \frac{\omega_j}{b_j}} \quad (1)$$

Then, WHM is called a weighted harmonic mean, where b_j ($j = 1, 2, \dots, n$) is a vector of positive real numbers, $\omega = (\omega_1, \omega_2, \dots, \omega_n)^T$ is the weight vector of b_j ($j = 1, 2, \dots, n$), with $\omega_j \geq 0$, and $\sum_{j=1}^n \omega_j = 1$. R^+ is the collection of all positive real numbers. Particularly, if $\omega_i = 1, \omega_j = 0, j \neq i$, then $WHM(b_1, b_2, \dots, b_n) = b_i$; if $\omega = (\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n})^T$, then the WHM is redefined as the harmonic mean (HM):

$$HM(b_1, b_2, \dots, b_n) = \frac{n}{\sum_{j=1}^n \frac{1}{b_j}} \quad (2)$$

Definition 4. Let $WHM : R^n \rightarrow R$ and b_j ($j = 1, 2, \dots, n$) be a vector of real numbers. If $\exists b_i = 0 (i = 1, 2, \dots, n_0)$, then the harmonic mean (HM) operator with zero values is defined as:

$$HM(b_1, b_2, \dots, b_n) = \left(\frac{n - n_0}{\sum_{j=1}^{n-n_0} \frac{1}{b_j}} \right) \times \left(\frac{n - n_0}{n} \right) \quad (3)$$

where b_j ($j = 1, 2, \dots, n - n_0$) is a nonzero value, n is the (total) sample size, and n_0 is the number of zero values.

Definition 5. [33] Let $\tilde{b}_j = (b_j^1, b_j^2, b_j^3)$ ($j = 1, 2, \dots, n$) be a vector of triangular fuzzy numbers and let $WHM : \psi^n \rightarrow \psi$, (ψ is the set of all triangular fuzzy numbers); then, a fuzzy weighted harmonic mean (FWHM) is defined as:

$$FWHM(\tilde{b}_1, \tilde{b}_2, \dots, \tilde{b}_n) = \frac{1}{\sum_{j=1}^n \frac{\omega_j}{b_j}} = \left(\frac{1}{\sum_{j=1}^n \frac{\omega_j}{b_j^1}}, \frac{1}{\sum_{j=1}^n \frac{\omega_j}{b_j^2}}, \frac{1}{\sum_{j=1}^n \frac{\omega_j}{b_j^3}} \right)$$

where $\omega = (\omega_1, \omega_2, \dots, \omega_n)^T$ is the weight vector of $\tilde{b}_j$, ($j = 1, 2, \dots, n$), $\omega_j \geq 0$, and $\sum_{j=1}^n \omega_j = 1$. Particularly, if $\omega_i = 1, \omega_j = 0, j \neq i$, then $FWHM(\tilde{b}_1, \tilde{b}_2, \dots, \tilde{b}_n) = \tilde{b}_i$; if $\omega = (\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n})^T$, then the FWHM is redefined as the fuzzy harmonic mean (FHM):

$$FHM(\tilde{b}_1, \tilde{b}_2, \dots, \tilde{b}_n) = \left(\frac{n}{\sum_{j=1}^n \frac{1}{b_j^1}}, \frac{n}{\sum_{j=1}^n \frac{1}{b_j^2}}, \frac{n}{\sum_{j=1}^n \frac{1}{b_j^3}} \right) \quad (5)$$

Definition 6. Let $\tilde{b}_j = (b_j^1, b_j^2, b_j^3)$ ($j = 1, 2, \dots, n$) be a vector of triangular fuzzy numbers, and let $FWHM : \psi^n \rightarrow \psi$. If $\exists b_i^\ell = 0 (i = 1, 2, \dots, n_0^\ell; \ell = 1, 2, 3)$, then the fuzzy harmonic mean (FHM) operator with zero values is defined as:

$$FHM(\tilde{b}_1, \tilde{b}_2, \dots, \tilde{b}_n) = \left(\frac{n - n_0^1}{\sum_{j=1}^{n-n_0^1} \frac{1}{b_j^1}}, \frac{n - n_0^2}{\sum_{j=1}^{n-n_0^2} \frac{1}{b_j^2}}, \frac{n - n_0^3}{\sum_{j=1}^{n-n_0^3} \frac{1}{b_j^3}} \right) \quad (6)$$

where b_j^ℓ ($j = 1, 2, \dots, n - n_0^\ell; \ell = 1, 2, 3$) is a nonzero value, n is the (total) sample size and n_0^ℓ is the number of zero values.

3. Methodology

3.1. Formulation of the FFMMLP problems

Let us consider the following FFMMLP problems:

The first-level decision-maker (FLDM):

$$\max_{\tilde{x}_1} \tilde{f}_1(\tilde{x}) = \max_{\tilde{x}_1} (\tilde{f}_{11}(\tilde{x}), \tilde{f}_{12}(\tilde{x}), \dots, \tilde{f}_{1s_1}(\tilde{x})),$$

$$\min_{\tilde{x}_1} \tilde{G}_1(\tilde{x}) = \min_{\tilde{x}_1} (\tilde{g}_{11}(\tilde{x}), \tilde{g}_{12}(\tilde{x}), \dots, \tilde{g}_{1m_1}(\tilde{x})),$$

where

$$\tilde{x}_2, \tilde{x}_3, \dots, \tilde{x}_n \text{ solves,} \quad (7)$$

The second-level decision-maker (SLDM):

$$\max_{\tilde{x}_2} \tilde{F}_2(\tilde{x}) = \max_{\tilde{x}_2} (\tilde{f}_{21}(\tilde{x}), \tilde{f}_{22}(\tilde{x}), \dots, \tilde{f}_{2S_2}(\tilde{x})),$$

$$\min_{\tilde{x}_2} \tilde{G}_2(\tilde{x}) = \min_{\tilde{x}_2} (\tilde{g}_{21}(\tilde{x}), \tilde{g}_{22}(\tilde{x}), \dots, \tilde{g}_{2m_2}(\tilde{x})),$$

where

$$\tilde{x}_3, \tilde{x}_4, \dots, \tilde{x}_n \text{ solves,} \quad (8)$$

:

where

$$\tilde{x}_p, \tilde{x}_{p+1}, \dots, \tilde{x}_n \text{ solves,}$$

*P*th-level decision-maker (*P*th LDM):

$$\max_{\tilde{x}_p} \tilde{F}_p(\tilde{x}) = \max_{\tilde{x}_p} (\tilde{f}_{p1}(\tilde{x}), \tilde{f}_{p2}(\tilde{x}), \dots, \tilde{f}_{pS_p}(\tilde{x})),$$

$$\min_{\tilde{x}_p} \tilde{G}_p(\tilde{x}) = \min_{\tilde{x}_p} (\tilde{g}_{p1}(\tilde{x}), \tilde{g}_{p2}(\tilde{x}), \dots, \tilde{g}_{pm_p}(\tilde{x})),$$

where

$$\tilde{x}_{p+1}, \tilde{x}_{p+2}, \dots, \tilde{x}_n \text{ solves,} \quad (9)$$

subject to

$$\tilde{G} = \left\{ \tilde{x} \in R^n \mid \sum_{j=1}^n \tilde{a}_{rj} \otimes \tilde{x}_j \leq \tilde{b}_r, \quad (r = 1, 2, \dots, k) \right. \\ \left. \tilde{x}_j \geq \tilde{0} \text{ for all } j. \right\} \quad (10)$$

where $\tilde{x}_i \in R^{n_i}, n_i \geq 1, (i = 1, 2, \dots, p)$ is a vector of fuzzy variables indicating the control of each level of decision-makers (DMs), $\sum_{i=1}^p n_i = n$. $\tilde{a}_{rj}$ is an $r \times j$ fuzzy-coefficient matrix, $\tilde{b}_r (r = 1, 2, \dots, k)$ is a fuzzy vector of TFNs, and $\tilde{F}_i(\tilde{x}) = (\tilde{f}_{i1}(\tilde{x}), \tilde{f}_{i2}(\tilde{x}), \dots, \tilde{f}_{iS_i}(\tilde{x}))$ and $\tilde{G}_i(\tilde{x}) = (\tilde{g}_{i1}(\tilde{x}), \tilde{g}_{i2}(\tilde{x}), \dots, \tilde{g}_{im_i}(\tilde{x}))$ ($i = 1, 2, \dots, p$) are the vectors of S_i and m_i distinct fuzzy objective functions for the i th-level decision-maker,

$$\tilde{f}_{it}(\tilde{x}) = \sum_{j=1}^n \tilde{c}_{ij}^t \otimes \tilde{x}_j \quad (i = 1, 2, \dots, p; \quad t = 1, 2, \dots, S_i)$$

$$\tilde{g}_{iq}(\tilde{x}) = \sum_{j=1}^n \tilde{d}_{ij}^q \otimes \tilde{x}_j \quad (i = 1, 2, \dots, p; \quad q = 1, 2, \dots, m_i) \quad (11)$$

$t = 1, 2, \dots, S_1$ and $q = 1, 2, \dots, m_1$ for the first-level objective functions, $t = 1, 2, \dots, S_2$ and $q = 1, 2, \dots, m_2$ for the second-level objective functions,

:

$t = 1, 2, \dots, S_p$ and $q = 1, 2, \dots, m_p$ for the *P*th-level objective functions.

3.2. Defuzzification process

Based on a crisp linear technique introduced by Jayalakshmi and Pandian [14] to treat fully fuzzy linear programming problems, FFMMLP problems can be converted into three crisp multiobjective linear programming (CMLP) problems for each level, and the three CMLP problems are treated one at a time.

Let all the fuzzy decision variables and fuzzy parameters in Eqs. (7)–(11) be presented by triangular fuzzy numbers, each of which has values as follows:

$$\tilde{F}_i = (v_i^1, v_i^2, v_i^3), \quad \tilde{G}_i = (h_i^1, h_i^2, h_i^3), \quad \tilde{f}_{it} = (u_{it}^1, u_{it}^2, u_{it}^3), \\ \tilde{g}_{iq} = (w_{iq}^1, w_{iq}^2, w_{iq}^3), \quad \tilde{c}_{ij}^t = (\alpha_{ij}^t, \beta_{ij}^t, \gamma_{ij}^t), \quad \tilde{d}_{ij}^q = (\delta_{ij}^q, \lambda_{ij}^q, \mu_{ij}^q), \\ \tilde{x}_j = (x_j, y_j, z_j), \quad \tilde{a}_{rj} = (a_{rj}^1, a_{rj}^2, a_{rj}^3) \text{ and } \tilde{b}_r = (b_r^1, b_r^2, b_r^3).$$

Hence, Eqs. (7)–(11) can be reformulated as follows:

$$[FLDM]: \max_{(x_1, y_1, z_1)} (v_1^1, v_1^2, v_1^3) \\ = \max_{(x_1, y_1, z_1)} ((u_{11}^1, u_{11}^2, u_{11}^3), (u_{12}^1, u_{12}^2, u_{12}^3), \dots, (u_{1S_1}^1, u_{1S_1}^2, u_{1S_1}^3)) \\ \min_{(x_1, y_1, z_1)} (h_1^1, h_1^2, h_1^3) = \min_{(x_1, y_1, z_1)} ((w_{11}^1, w_{11}^2, w_{11}^3), (w_{12}^1, w_{12}^2, w_{12}^3), \dots, (w_{1m_1}^1, w_{1m_1}^2, w_{1m_1}^3)) \\ \text{where } (x_2, y_2, z_2), (x_3, y_3, z_3), \dots, (x_n, y_n, z_n) \text{ solves,} \quad (12)$$

$$[SLDM]: \max_{(x_2, y_2, z_2)} (v_2^1, v_2^2, v_2^3) \\ = \max_{(x_2, y_2, z_2)} ((u_{21}^1, u_{21}^2, u_{21}^3), (u_{22}^1, u_{22}^2, u_{22}^3), \dots, (u_{2S_2}^1, u_{2S_2}^2, u_{2S_2}^3)) \\ \min_{(x_2, y_2, z_2)} (h_2^1, h_2^2, h_2^3) = \min_{(x_2, y_2, z_2)} ((w_{21}^1, w_{21}^2, w_{21}^3), (w_{22}^1, w_{22}^2, w_{22}^3), \dots, (w_{2m_2}^1, w_{2m_2}^2, w_{2m_2}^3)) \\ \text{where } (x_3, y_3, z_3), (x_4, y_4, z_4), \dots, (x_n, y_n, z_n) \text{ solves,} \quad (13)$$

:

$$\text{where } (x_p, y_p, z_p), (x_{p+1}, y_{p+1}, z_{p+1}), \dots, (x_n, y_n, z_n) \text{ solves,} \\ [PthLDM]: \max_{(x_p, y_p, z_p)} (v_p^1, v_p^2, v_p^3) \\ = \max_{(x_p, y_p, z_p)} ((u_{p1}^1, u_{p1}^2, u_{p1}^3), (u_{p2}^1, u_{p2}^2, u_{p2}^3), \dots, (u_{pS_p}^1, u_{pS_p}^2, u_{pS_p}^3)) \\ \min_{(x_p, y_p, z_p)} (h_p^1, h_p^2, h_p^3) = \min_{(x_p, y_p, z_p)} ((w_{p1}^1, w_{p1}^2, w_{p1}^3), (w_{p2}^1, w_{p2}^2, w_{p2}^3), \dots, (w_{pm_p}^1, w_{pm_p}^2, w_{pm_p}^3)) \\ \text{where } (x_{p+1}, y_{p+1}, z_{p+1}), \dots, (x_n, y_n, z_n) \text{ solves,} \quad (14)$$

subject to

$$x \in G = \left\{ x \in R^n \mid \sum_{j=1}^n (a_{rj}^1, a_{rj}^2, a_{rj}^3) \otimes (x_j, y_j, z_j) \leq (b_r^1, b_r^2, b_r^3), (r = 1, 2, \dots, k) \right. \\ \left. x_j, y_j, z_j \geq 0 \text{ for all } j. \right\} \quad (15)$$

and the objective functions of the i th-level decision-maker in crisp form are obtained as follows:

$$f_{it}(x_j, y_j, z_j) = (u_{it}^1, u_{it}^2, u_{it}^3) = \sum_{j=1}^n (\alpha_{ij}^t, \beta_{ij}^t, \gamma_{ij}^t) \otimes (x_j, y_j, z_j)$$

$$(i = 1, 2, \dots, p; \quad t = 1, 2, \dots, S_i)$$

$$g_{iq}(x_j, y_j, z_j) = (w_{iq}^1, w_{iq}^2, w_{iq}^3) = \sum_{j=1}^n (\delta_{ij}^q, \lambda_{ij}^q, \mu_{ij}^q) \otimes (x_j, y_j, z_j)$$

$$(i = 1, 2, \dots, p; q = 1, 2, \dots, m_i) \quad (16)$$

Now, using the arithmetic operations of Definition 1 and the partial ordering relations described in Definition 2, Eqs. (12)–(16) can be rewritten in the following crisp multiobjective linear programming problems for each level:

[ith level decision – maker] ($i = 1, 2, \dots, p$) :

$$\max_{x_i} v_i^1 = \begin{cases} \max_{x_i} u_{i1}^1 = \sum_{j=1}^n \text{lower value of } ((\alpha_{ij}^1, \beta_{ij}^1, \gamma_{ij}^1) \otimes (x_j, y_j, z_j)), \\ \max_{x_i} u_{i2}^1 = \sum_{j=1}^n \text{lower value of } ((\alpha_{ij}^2, \beta_{ij}^2, \gamma_{ij}^2) \otimes (x_j, y_j, z_j)), \\ \vdots \\ \max_{x_i} u_{iS_i}^1 = \sum_{j=1}^n \text{lower value of } ((\alpha_{ij}^{S_i}, \beta_{ij}^{S_i}, \gamma_{ij}^{S_i}) \otimes (x_j, y_j, z_j)), \end{cases} \quad (17)$$

$$\min_{x_i} h_i^1 = \begin{cases} \min_{x_i} w_{i1}^1 = \sum_{j=1}^n \text{lower value of } ((\delta_{ij}^1, \lambda_{ij}^1, \mu_{ij}^1) \otimes (x_j, y_j, z_j)), \\ \min_{x_i} w_{i2}^1 = \sum_{j=1}^n \text{lower value of } ((\delta_{ij}^2, \lambda_{ij}^2, \mu_{ij}^2) \otimes (x_j, y_j, z_j)), \\ \vdots \\ \min_{x_i} w_{iS_i}^1 = \sum_{j=1}^n \text{lower value of } ((\delta_{ij}^{S_i}, \lambda_{ij}^{S_i}, \mu_{ij}^{S_i}) \otimes (x_j, y_j, z_j)), \end{cases} \quad (18)$$

$$\max_{y_i} v_i^2 = \begin{cases} \max_{y_i} u_{i1}^2 = \sum_{j=1}^n \text{middle value of } ((\alpha_{ij}^1, \beta_{ij}^1, \gamma_{ij}^1) \otimes (x_j, y_j, z_j)), \\ \max_{y_i} u_{i2}^2 = \sum_{j=1}^n \text{middle value of } ((\alpha_{ij}^2, \beta_{ij}^2, \gamma_{ij}^2) \otimes (x_j, y_j, z_j)), \\ \vdots \\ \max_{y_i} u_{iS_i}^2 = \sum_{j=1}^n \text{middle value of } ((\alpha_{ij}^{S_i}, \beta_{ij}^{S_i}, \gamma_{ij}^{S_i}) \otimes (x_j, y_j, z_j)), \end{cases} \quad (19)$$

$$\min_{y_i} h_i^2 = \begin{cases} \min_{y_i} w_{i1}^2 = \sum_{j=1}^n \text{middle value of } ((\delta_{ij}^1, \lambda_{ij}^1, \mu_{ij}^1) \otimes (x_j, y_j, z_j)), \\ \min_{y_i} w_{i2}^2 = \sum_{j=1}^n \text{middle value of } ((\delta_{ij}^2, \lambda_{ij}^2, \mu_{ij}^2) \otimes (x_j, y_j, z_j)), \\ \vdots \\ \min_{y_i} w_{iS_i}^2 = \sum_{j=1}^n \text{middle value of } ((\delta_{ij}^{S_i}, \lambda_{ij}^{S_i}, \mu_{ij}^{S_i}) \otimes (x_j, y_j, z_j)), \end{cases} \quad (20)$$

$$\max_{z_i} v_i^3 = \begin{cases} \max_{z_i} u_{i1}^3 = \sum_{j=1}^n \text{upper value of } ((\alpha_{ij}^1, \beta_{ij}^1, \gamma_{ij}^1) \otimes (x_j, y_j, z_j)), \\ \max_{z_i} u_{i2}^3 = \sum_{j=1}^n \text{upper value of } ((\alpha_{ij}^2, \beta_{ij}^2, \gamma_{ij}^2) \otimes (x_j, y_j, z_j)), \\ \vdots \\ \max_{z_i} u_{iS_i}^3 = \sum_{j=1}^n \text{upper value of } ((\alpha_{ij}^{S_i}, \beta_{ij}^{S_i}, \gamma_{ij}^{S_i}) \otimes (x_j, y_j, z_j)), \end{cases} \quad (21)$$

$$\min_{z_i} h_i^3 = \begin{cases} \min_{z_i} w_{i1}^3 = \sum_{j=1}^n \text{upper value of } ((\delta_{ij}^1, \lambda_{ij}^1, \mu_{ij}^1) \otimes (x_j, y_j, z_j)), \\ \min_{z_i} w_{i2}^3 = \sum_{j=1}^n \text{upper value of } ((\delta_{ij}^2, \lambda_{ij}^2, \mu_{ij}^2) \otimes (x_j, y_j, z_j)), \\ \vdots \\ \min_{z_i} w_{iS_i}^3 = \sum_{j=1}^n \text{upper value of } ((\delta_{ij}^{S_i}, \lambda_{ij}^{S_i}, \mu_{ij}^{S_i}) \otimes (x_j, y_j, z_j)), \end{cases} \quad (22)$$

where $(x_{i+1}, y_{i+1}, z_{i+1}), \dots, (x_n, y_n, z_n)$ solves

subject to

$$x \in G_d = \begin{cases} \sum_{j=1}^n \text{lower value of } ((a_{ij}^1, a_{ij}^2, a_{ij}^3) \otimes (x_j, y_j, z_j)) \leq b_i^1, \\ \sum_{j=1}^n \text{middle value of } ((a_{ij}^1, a_{ij}^2, a_{ij}^3) \otimes (x_j, y_j, z_j)) \leq b_i^2, \\ \sum_{j=1}^n \text{upper value of } ((a_{ij}^1, a_{ij}^2, a_{ij}^3) \otimes (x_j, y_j, z_j)) \leq b_i^3, \\ u_{it}^2 \geq u_{it}^1, u_{it}^3 \geq u_{it}^2 (t = 1, 2, \dots, S_i), \\ y_j \geq x_j, z_j \geq y_j, x_j \geq 0 \text{ for all } j. \end{cases} \quad (23)$$

3.3. Multilevel programming problems with a single objective function

3.3.1. First-Level Decision-Maker (FLDM) problems

Find a single optimum solution that corresponds to each of the decomposed objective functions of FLDM problems, as follows:

$$u_{1t}^L = \max_{x \in G_d} u_{1t}^1, \quad u_{1t}^M = \max_{x \in G_d} u_{1t}^2, \quad \text{and} \\ u_{1t}^U = \max_{x \in G_d} u_{1t}^3 \quad (t = 1, 2, \dots, S_1). \quad (24)$$

$$w_{1q}^L = \min_{x \in G_d} w_{1q}^1, \quad w_{1q}^M = \min_{x \in G_d} w_{1q}^2, \quad \text{and} \\ w_{1q}^U = \min_{x \in G_d} w_{1q}^3 \quad (t = 1, 2, \dots, m_1). \quad (25)$$

where $u_{1t}^L, u_{1t}^M, u_{1t}^U, w_{1q}^L, w_{1q}^M, w_{1q}^U$ ($t = 1, 2, \dots, S_1; q = 1, 2, \dots, m_1$) are the individual optimum solutions for each objective function of FLDM problems.

Calculate the fuzzy harmonic means FHM_1^u and FHM_1^w of the optimum values of the decomposed objective functions u_{1t}^ℓ and w_{1q}^ℓ , respectively ($t = 1, 2, \dots, S_1; q = 1, 2, \dots, m_1$) ($\ell = 1, 2, 3$) of FLDM problems.

$$FHM_1^u = (HM_{1u}^L, HM_{1u}^M, HM_{1u}^U) \\ = \left(\frac{S_1}{\sum_{t=1}^{S_1} \left(\frac{1}{u_{1t}^L} \right)}, \frac{S_1}{\sum_{t=1}^{S_1} \left(\frac{1}{u_{1t}^M} \right)}, \frac{S_1}{\sum_{t=1}^{S_1} \left(\frac{1}{u_{1t}^U} \right)} \right) \quad (26)$$

$$FHM_1^w = (HM_{1w}^L, HM_{1w}^M, HM_{1w}^U) \\ = \left(\frac{m_1}{\sum_{q=1}^{m_1} \left(\frac{1}{w_{1q}^L} \right)}, \frac{m_1}{\sum_{q=1}^{m_1} \left(\frac{1}{w_{1q}^M} \right)}, \frac{m_1}{\sum_{q=1}^{m_1} \left(\frac{1}{w_{1q}^U} \right)} \right) \quad (27)$$

Crisp multiobjective linear programming (CMLP₁) problems for the FLDM can be transformed into three crisp single objective linear programming (CSLP₁) problems, as follows:

[FLDM]:

$$L - CSLP_1 : \max_{x_1} \lambda_1^1 \\ = \frac{\sum_{t=1}^{S_1} (\max u_{1t}^1)}{HM_{1u}^U} - \frac{\sum_{q=1}^{m_1} (\min w_{1q}^1)}{HM_{1w}^U} \quad (28)$$

$$M - CSLP_1 : \max_{y_1} \lambda_1^2 \\ = \frac{\sum_{t=1}^{S_1} (\max u_{1t}^2)}{HM_{1u}^M} - \frac{\sum_{q=1}^{m_1} (\min w_{1q}^2)}{HM_{1w}^M} \quad (29)$$

$$U - CSLP_1 : \max_{z_1} \lambda_1^3 \\ = \frac{\sum_{t=1}^{S_1} (\max u_{1t}^3)}{HM_{1u}^L} - \frac{\sum_{q=1}^{m_1} (\min w_{1q}^3)}{HM_{1w}^L} \quad (30)$$

where

$(x_2, y_2, z_2), \dots, (x_n, y_n, z_n)$ solves,

subject to

$$x \in G_d,$$

$$\lambda_1^1 \leq \lambda_1^2, \lambda_1^2 \leq \lambda_1^3. \quad (31)$$

Find the individual optimal solutions of the objective functions λ_1^ℓ ($\ell = 1, 2, 3$) of the FLDM problems, as follows:

$$\lambda_F^\ell = \max_{x \in G_d} \lambda_1^\ell \quad (\ell = 1, 2, 3) \quad (32)$$

Calculating the harmonic mean (HM^F) of the optimum values of the objective functions λ_1^ℓ ($\ell = 1, 2, 3$), we obtain

$$HM^F = \frac{3}{\sum_{\ell=1}^3 \left(\frac{1}{\lambda_F^\ell} \right)} \quad (33)$$

The equivalent single objective function of the FLDM problems using the harmonic mean technique [29] can be written as follows:

$$SLP_1 : \max_{(x_1, y_1, z_1)} \xi_1 = \frac{\sum_{\ell=1}^3 (\max \lambda_1^\ell)}{HM^F} \quad (34)$$

where $(x_2, y_2, z_2), \dots, (x_n, y_n, z_n)$ solves,

subject to

$$x \in G_d,$$

$$\lambda_1^1 \leq \lambda_1^2, \lambda_1^2 \leq \lambda_1^3. \quad (35)$$

The efficient fuzzy solution to FLDM problems is $(\tilde{x}_1^{1st}, \tilde{x}_2^{1st}, \dots, \tilde{x}_n^{1st})$.

3.3.2. Second-Level Decision-Maker (SLDM) problems

Find a single optimum solution that corresponds to each of the decomposed objective functions of SLDM problems, as follows:

$$u_{2t}^L = \max_{\substack{x \in G_d \\ \tilde{x}_1 = \tilde{x}_1^{1st}}} u_{2t}^1, \quad u_{2t}^M = \max_{\substack{x \in G_d \\ \tilde{x}_1 = \tilde{x}_1^{1st}}} u_{2t}^2, \quad \text{and} \quad u_{2t}^U = \max_{\substack{x \in G_d \\ \tilde{x}_1 = \tilde{x}_1^{1st}}} u_{2t}^3 \quad (t = 1, 2, \dots, S_2) \quad (36)$$

$$w_{2q}^L = \min_{\substack{x \in G_d \\ \tilde{x}_1 = \tilde{x}_1^{1st}}} w_{2q}^1, \quad w_{2q}^M = \min_{\substack{x \in G_d \\ \tilde{x}_1 = \tilde{x}_1^{1st}}} w_{2q}^2, \quad \text{and} \quad w_{2q}^U = \min_{\substack{x \in G_d \\ \tilde{x}_1 = \tilde{x}_1^{1st}}} w_{2q}^3 \quad (q = 1, 2, \dots, m_2) \quad (37)$$

where $u_{2t}^L, u_{2t}^M, u_{2t}^U, w_{2q}^L, w_{2q}^M$ and w_{2q}^U ($t = 1, 2, \dots, S_2; q = 1, 2, \dots, m_2$) are the individual optimum solutions for each objective function of SLDM problems.

Calculate the fuzzy harmonic means FHM_2^u and FHM_2^w of the optimum values of the decomposed objective functions u_{2t}^ℓ and w_{2q}^ℓ , respectively ($t = 1, 2, \dots, S_2; q = 1, 2, \dots, m_2$) ($\ell = 1, 2, 3$) of SLDM problems.

$$FHM_2^u = (HM_{2u}^L, HM_{2u}^M, HM_{2u}^U) \\ = \left(\frac{S_2}{\sum_{t=1}^{S_2} \left(\frac{1}{u_{2t}^L} \right)}, \frac{S_2}{\sum_{t=1}^{S_2} \left(\frac{1}{u_{2t}^M} \right)}, \frac{S_2}{\sum_{t=1}^{S_2} \left(\frac{1}{u_{2t}^U} \right)} \right) \quad (38)$$

$$FHM_2^w = (HM_{2w}^L, HM_{2w}^M, HM_{2w}^U) \\ = \left(\frac{m_2}{\sum_{q=1}^{m_2} \left(\frac{1}{w_{2q}^L} \right)}, \frac{m_2}{\sum_{q=1}^{m_2} \left(\frac{1}{w_{2q}^M} \right)}, \frac{m_2}{\sum_{q=1}^{m_2} \left(\frac{1}{w_{2q}^U} \right)} \right) \quad (39)$$

Using the FLDM variable $\tilde{x}_1^{1st}$ in SLDM problems, the crisp multiobjective linear programming (CMLP₂) problems can be transformed into three crisp single-objective linear programming (CSLP₂) problems, as follows:

[SLDM]:

$$L - CSLP_2 : \max_{x_2} \lambda_2^1 \\ = \frac{\sum_{t=1}^{S_2} (\max u_{2t}^1)}{HM_{2u}^U} - \frac{\sum_{q=1}^{m_2} (\min w_{2q}^1)}{HM_{2w}^U}, \quad (40)$$

$$M - CSLP_2 : \max_{y_2} \lambda_2^2 \\ = \frac{\sum_{t=1}^{S_2} (\max u_{2t}^2)}{HM_{2u}^M} - \frac{\sum_{q=1}^{m_2} (\min w_{2q}^2)}{HM_{2w}^M}, \quad (41)$$

$$U - CSLP_2 : \max_{z_2} \lambda_2^3 \\ = \frac{\sum_{t=1}^{S_2} (\max u_{2t}^3)}{HM_{2u}^L} - \frac{\sum_{q=1}^{m_2} (\min w_{2q}^3)}{HM_{2w}^L}, \quad (42)$$

where

$(x_3, y_3, z_3), \dots, (x_n, y_n, z_n)$ solves,

subject to

$$x \in G_d,$$

$$\lambda_2^1 \leq \lambda_2^2, \lambda_2^2 \leq \lambda_2^3,$$

$$x_1 = x_1^{1st}, y_1 = y_1^{1st}, z_1 = z_1^{1st} \quad (43)$$

Find the individual optimal solutions of the objective functions λ_2^ℓ ($\ell = 1, 2, 3$) of the SLDM problems, as follows:

$$\lambda_S^\ell = \max_{x \in G_d} \lambda_2^\ell \quad (\ell = 1, 2, 3) \quad (44)$$

Calculating the harmonic mean (HM^S) of the optimum values of the objectives λ_2^ℓ ($\ell = 1, 2, 3$), we obtain

$$HM^S = \frac{3}{\sum_{\ell=1}^3 \left(\frac{1}{\lambda_S^\ell} \right)} \quad (45)$$

The equivalent single objective function of the SLDM problems using the harmonic mean technique [29] can be written as follows:

$$SLP_2 : \max_{(x_2, y_2, z_2)} \xi_2 = \frac{\sum_{\ell=1}^3 (\max \lambda_2^\ell)}{HM^S} \quad (46)$$

where $(x_3, y_3, z_3), \dots, (x_n, y_n, z_n)$ solves,

subject to

$$x \in G_d,$$

$$\lambda_2^1 \leq \lambda_2^2, \lambda_2^2 \leq \lambda_2^3,$$

$$x_1 = x_1^{1st}, y_1 = y_1^{1st}, z_1 = z_1^{1st} \quad (47)$$

The efficient fuzzy solution to SLDM problems is $(\tilde{x}_1^{1st}, \tilde{x}_2^{2nd}, \dots, \tilde{x}_n^{2nd})$.

3.3.3. Pth-Level Decision Maker (Pth-LDM) problems

Find a single optimum solution that corresponds to each of the decomposed objective functions of the Pth-LDM problems, as follows:

$$u_{pt}^L = \max_{\substack{x \in G_d \\ \tilde{x}_j = \tilde{x}_j^{ith}}} u_{pt}^1, \quad u_{pt}^M = \max_{\substack{x \in G_d \\ \tilde{x}_j = \tilde{x}_j^{ith}}} u_{pt}^2, \quad \text{and} \quad u_{pt}^U = \max_{\substack{x \in G_d \\ \tilde{x}_j = \tilde{x}_j^{ith}}} u_{pt}^3 \quad (i, j = 1, 2, \dots, p-1) (t = 1, 2, \dots, S_p) \quad (48)$$

$$w_{pq}^L = \min_{\substack{x \in G_d \\ \tilde{x}_j = \tilde{x}_j^{ith}}} w_{pq}^1, \quad w_{pq}^M = \min_{\substack{x \in G_d \\ \tilde{x}_j = \tilde{x}_j^{ith}}} w_{pq}^2, \quad \text{and} \quad w_{pq}^U = \min_{\substack{x \in G_d \\ \tilde{x}_j = \tilde{x}_j^{ith}}} w_{pq}^3 \quad (i, j = 1, 2, \dots, p-1) (q = 1, 2, \dots, m_p) \quad (49)$$

where $u_{pt}^L, u_{pt}^M, u_{pt}^U, w_{pq}^L, w_{pq}^M$ and w_{pq}^U ($t = 1, 2, \dots, S_p; q = 1, 2, \dots, m_p$) are the individual optimum solutions for each objective function of the Pth-LDM problems.

Calculate the fuzzy harmonic means FHM_p^u and FHM_p^w of the optimum values of the decomposed objective functions u_{pt}^ℓ and w_{pq}^ℓ , respectively ($t = 1, 2, \dots, S_p; q = 1, 2, \dots, m_p$) ($\ell = 1, 2, 3$) of the Pth-LDM problems.

$$FHM_p^u = (HM_{pu}^L, HM_{pu}^M, HM_{pu}^U) = \left(\frac{S_p}{\sum_{t=1}^{S_p} \left(\frac{1}{u_{pt}^L} \right)}, \frac{S_p}{\sum_{t=1}^{S_p} \left(\frac{1}{u_{pt}^M} \right)}, \frac{S_p}{\sum_{t=1}^{S_p} \left(\frac{1}{u_{pt}^U} \right)} \right) \quad (50)$$

$$FHM_p^w = (HM_{pw}^L, HM_{pw}^M, HM_{pw}^U) = \left(\frac{m_p}{\sum_{q=1}^{m_p} \left(\frac{1}{w_{pq}^L} \right)}, \frac{m_p}{\sum_{q=1}^{m_p} \left(\frac{1}{w_{pq}^M} \right)}, \frac{m_p}{\sum_{q=1}^{m_p} \left(\frac{1}{w_{pq}^U} \right)} \right) \quad (51)$$

Using the i th-level decision-maker (ith-LDM) variable $\tilde{x}_j^{ith}$ ($i, j = 1, 2, \dots, p-1$) in the Pth-LDM problems, the crisp multiobjective linear programming (CMLP_p) problems can be transformed into three crisp single-objective linear programming (CSLP_p) problems, as follows:

[Pth-LDM]:

$$L - CSLP_p : \max_{x_p} \lambda_p^1 = \frac{\sum_{t=1}^{S_p} \left(\max u_{pt}^1 \right)}{HM_{pu}^U} - \frac{\sum_{q=1}^{m_p} \left(\min w_{pq}^1 \right)}{HM_{pw}^U}, \quad (52)$$

$$M - CSLP_p : \max_{y_p} \lambda_p^2 = \frac{\sum_{t=1}^{S_p} \left(\max u_{pt}^2 \right)}{HM_{pu}^M} - \frac{\sum_{q=1}^{m_p} \left(\min w_{pq}^2 \right)}{HM_{pw}^M} \quad (53)$$

$$U - CSLP_p : \max_{z_p} \lambda_p^3 = \frac{\sum_{t=1}^{S_p} \left(\max u_{pt}^3 \right)}{HM_{pu}^L} - \frac{\sum_{q=1}^{m_p} \left(\min w_{pq}^3 \right)}{HM_{pw}^L} \quad (54)$$

where $(x_{p+1}, y_{p+1}, z_{p+1}), \dots, (x_n, y_n, z_n)$ solves,

subject to

$$x \in G_d,$$

$$\lambda_p^1 \leq \lambda_p^2, \lambda_p^2 \leq \lambda_p^3,$$

$$x_j = x_j^{ith}, y_j = y_j^{ith}, z_j = z_j^{ith} \quad (i, j = 1, 2, \dots, p-1) \quad (55)$$

Find the individual optimal solutions of the objective functions λ_p^ℓ ($\ell = 1, 2, 3$) of the Pth LDM problems, as follows:

$$\lambda_{pth}^\ell = \max_{x \in G_d} \lambda_p^\ell \quad (\ell = 1, 2, 3) \quad (56)$$

Calculating the harmonic mean (HM^{pth}) of the optimum values of the objectives λ_p^ℓ ($\ell = 1, 2, 3$), we obtain

$$HM^{pth} = \frac{3}{\sum_{\ell=1}^3 \left(\frac{1}{\lambda_{pth}^\ell} \right)} \quad (57)$$

The equivalent single objective function of the Pth LDM problems using the harmonic mean technique [29] can be written as follows:

$$SLP_p : \max_{(x_p, y_p, z_p)} \zeta_p = \frac{\sum_{\ell=1}^3 \left(\max \lambda_p^\ell \right)}{HM^{pth}} \quad (58)$$

where $(x_{p+1}, y_{p+1}, z_{p+1}), \dots, (x_n, y_n, z_n)$ solves,

subject to

$$x \in G_d,$$

$$\lambda_p^1 \leq \lambda_p^2, \lambda_p^2 \leq \lambda_p^3,$$

$$x_j = x_j^{ith}, y_j = y_j^{ith}, z_j = z_j^{ith} \quad (i, j = 1, 2, \dots, p-1) \quad (59)$$

Therefore, the fuzzy compromise solution to an FFMMLP problem is given by $\tilde{x}_j^*$ ($j = 1, 2, \dots, n$) and $\tilde{f}_{it}^*, \tilde{g}_{iq}^*$ ($q = 1, 2, \dots, m_i; t = 1, 2, \dots, S_i$) ($i = 1, 2, \dots, p$).

4. Procedure to solve FFMMLP problems

In this section, we introduce the steps of the procedures of solving FFMMLP problems as follows:

Step1:Formulate the FFMMLP problems as Eqs. (7)–(10)

Step2:Set $i = 1$.

Step3:If the i^{th} -LDM is satisfied, go to step 10. Otherwise, go to step 4.

Step4:Apply the crisp linear technique [14] and arithmetic operations to formulate the crisp model (17)–(23)

Step5:Calculate the individual optimal solution of the objective functions u_{it}^{ℓ} and w_{iq}^{ℓ} .

Step6:The results of step 5 determine the fuzzy harmonic means FHM_i^u and FHM_i^w .

Step7:Use the harmonic mean technique to construct three crisp single objective linear programming problems, $L - CSLP_i$, $M - CSLP_i$ and $U - CSLP_i$.

Step8:Evaluate the individual optimal solution of the objective functions λ_i^{ℓ} of Eqs. (28)–(30).

Step9:From the results of step 8, calculate the harmonic mean HM^i .

Step10:
Utilize the harmonic mean to formulate and solve the equivalent single objective linear programming problem SLP_i .

Step11:
If the decision-maker of the i^{th} -LDM problems is satisfied, go to step 13. Otherwise, go to step 12.

Step12:
All decision variables of the i^{th} -LDM problems must satisfy the bounded variables constraints.

Step13:
The i^{th} -LDM obtains the efficient fuzzy solution $(\tilde{x}_1^i, \tilde{x}_2^i, \dots, \tilde{x}_n^i)$ and proceeds to the next step.

Step14:
If $i = n$ go to step 18, otherwise go to step 15.

Step15:
Set $i = i + 1$ go to the next step.

Step16:
Define the i^{th} LDM's problems from the point of view of the $(i-1)^{\text{th}}$ LDM.

Step17:
Set $x_j = x_j^{ith}$, $y_j = y_j^{ith}$ and $z_j = z_j^{ith}$ go to Step 3.

Step18:
 $(\tilde{x}_{*1}^{(i-1)th}, \tilde{x}_{*2}^{(i-1)th}, \tilde{x}_{*3}^{(i-1)th}, \dots, \tilde{x}_{*n}^{(i-1)th})$ is an efficient fuzzy solution for the i^{th} LDM.

Step19:

The efficient fuzzy solution of the FFMMLP problems is obtained; go to step 20.

Step20:
Stop.

The decision-making procedure of the proposed problem is presented in Fig. 1.

5. Numerical studies

In this section, we present a numerical illustration of the FFMMLP problems for the methodology's effectiveness. Additionally, the proposed methodology is applied for the network product supply chain.

5.1. Numerical example

The following example demonstrates the computational procedure for an FFMMLP problem:

$$\text{First-level decision-maker [FLDM]} : \max_{\tilde{x}_1} (\tilde{f}_{11}, \tilde{f}_{12}, \tilde{f}_{13})$$

where $\tilde{x}_2, \tilde{x}_3$ solves,

$$\text{Second-level decision-maker [SLDM]} : \max_{\tilde{x}_2} (\tilde{f}_{21}, \tilde{f}_{22})$$

where $\tilde{x}_3$ solves,

$$\text{Third-level decision-maker [TLDM]} : \max_{\tilde{x}_3} (\tilde{f}_{31}, \tilde{f}_{32}, \tilde{f}_{33})$$

subject to

$$(2, 4, 5) \otimes \tilde{x}_1 \oplus (1, 5, 7) \otimes \tilde{x}_2 \oplus (3, 6, 8) \otimes \tilde{x}_3 \leq (30, 40, 50),$$

$$(3, 4, 6) \otimes \tilde{x}_1 \oplus (2, 5, 6) \otimes \tilde{x}_2 \oplus (4, 8, 9) \otimes \tilde{x}_3 \leq (15, 27, 42),$$

$$(5, 8, 12) \otimes \tilde{x}_1 \oplus (1, 3, 4) \otimes \tilde{x}_2 \oplus (6, 8, 11) \otimes \tilde{x}_3 \leq (20, 36, 58),$$

$$\tilde{x}_1, \tilde{x}_2, \tilde{x}_3 \geq \tilde{0}.$$

where

$$\tilde{f}_{11} = (6, 9, 13) \otimes \tilde{x}_1 \oplus (3, 4, 5) \otimes \tilde{x}_2 \oplus (4, 6, 7) \otimes \tilde{x}_3,$$

$$\tilde{f}_{12} = (8, 11, 14) \otimes \tilde{x}_1 \oplus (5, 7, 8) \otimes \tilde{x}_2 \oplus (2, 4, 7) \otimes \tilde{x}_3,$$

$$\tilde{f}_{13} = (10, 12, 15) \otimes \tilde{x}_1 \oplus (1, 3, 6) \otimes \tilde{x}_2 \oplus (3, 5, 7) \otimes \tilde{x}_3,$$

$$\tilde{f}_{21} = (12, 14, 17) \otimes \tilde{x}_1 \oplus (6, 7, 10) \otimes \tilde{x}_2 \oplus (4, 5, 8) \otimes \tilde{x}_3,$$

$$\tilde{f}_{22} = (9, 12, 16) \otimes \tilde{x}_1 \oplus (1, 2, 4) \otimes \tilde{x}_2 \oplus (3, 6, 8) \otimes \tilde{x}_3,$$

$$\tilde{f}_{31} = (1, 3, 6) \otimes \tilde{x}_1 \oplus (2, 7, 10) \otimes \tilde{x}_2 \oplus (5, 14, 16) \otimes \tilde{x}_3,$$

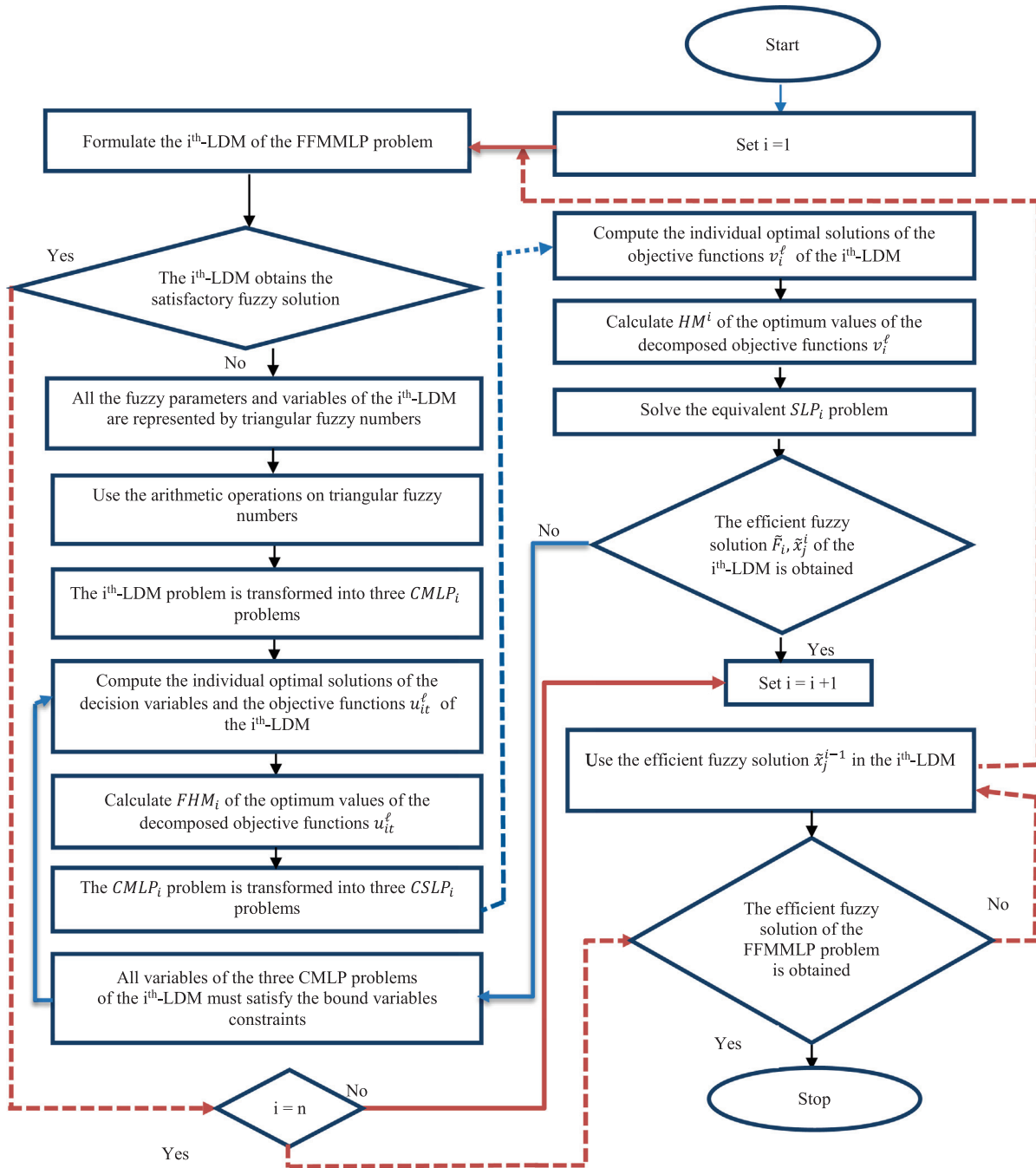


Fig. 1 A flowchart to illustrate the decision-making procedure of the proposed problem.

$$\tilde{f}_{32} = (2, 8, 10) \otimes \tilde{x}_1 \oplus (3, 9, 12) \otimes \tilde{x}_2 \oplus (7, 18, 20) \otimes \tilde{x}_3,$$

$$\tilde{f}_{33} = (3, 9, 11) \otimes \tilde{x}_1 \oplus (3, 11, 13) \otimes \tilde{x}_2 \oplus (10, 22, 25) \otimes \tilde{x}_3$$

The beginning step in the procedure is to use arithmetical operations and partial ordering relations [15,33] to transform an FFMMLP problem into a crisp multilevel multiobjective linear programming problem.

Let $\tilde{x}_j = (x_j, y_j, z_j)$ ($j = 1, 2, 3$) and $\tilde{f}_{it} = (u_{it}^1, u_{it}^2, u_{it}^3)$ ($i, t = 1, 2, 3$).

First, a crisp multiobjective linear programming problem for the FLDM, CMLP₁, can be modeled as follows:

[FLDM]:

$$\begin{aligned} \text{L - CMLP}_1 : \quad & \max_{x_1} (u_{11}^1, u_{12}^1, u_{13}^1) \\ & = (6x_1 + 3x_2 + 4x_3, 8x_1 + 5x_2 + 2x_3, 10x_1 + x_2 + 3x_3), \end{aligned}$$

$$\begin{aligned} \text{M - CMLP}_1 : \max_{y_1} (u_{11}^2, u_{12}^2, u_{13}^2) \\ = (9y_1 + 4y_2 + 6y_3, 11y_1 + 7y_2 + 4y_3, 12y_1 + 3y_2 + 5y_3), \end{aligned}$$

$$\begin{aligned} \text{U - CMLP}_1 : \max_{z_1} (u_{11}^3, u_{12}^3, u_{13}^3) \\ = (13z_1 + 5z_2 + 7z_3, 14z_1 + 8z_2 + 7z_3, 15z_1 + 6z_2 + 7z_3), \end{aligned}$$

where

$$(x_2, y_2, z_2), (x_3, y_3, z_3) \text{ solves,}$$

subject to

$$G_d = \{2x_1 + x_2 + 3x_3 \leq 30,$$

$$3x_1 + 2x_2 + 4x_3 \leq 15,$$

$$5x_1 + x_2 + 6x_3 \leq 20,$$

$$4y_1 + 5y_2 + 6y_3 \leq 40,$$

$$4y_1 + 5y_2 + 8y_3 \leq 27,$$

$$8y_1 + 3y_2 + 8y_3 \leq 36,$$

$$5z_1 + 7z_2 + 8z_3 \leq 50,$$

$$6z_1 + 6z_2 + 9z_3 \leq 42,$$

$$12z_1 + 4z_2 + 11z_3 \leq 58,$$

$$x_j \leq y_j, y_j \leq z_j (j = 1, 2, 3),$$

$$u_{it}^1 \leq u_{it}^2, u_{it}^2 \leq u_{it}^3 (i, t = 1, 2, 3),$$

$$x_j, y_j, z_j \geq 0 (j = 1, 2, 3). \}$$

The individual optimal solutions for the decision variables (x_j, y_j, z_j) ($j = 1, 2, 3$) and the objective functions u_{it}^{ℓ} ($t, \ell = 1, 2, 3$) of an FLDM problem are given in Table 1.

Calculating the fuzzy harmonic mean FHM_1 of the optimum values of the decomposed objective functions u_{it}^{ℓ} ($t, \ell = 1, 2, 3$), we have

$$FHM_1 = (HM_1^L, HM_1^M, HM_1^U) = (34.7415, 50.0308, 71.8926)$$

A crisp multiobjective linear programming CMLP₁ problem for an FLDM problem can be transformed into a crisp single-objective linear programming CSLP₁ problem as follows:

[FLDM]:

$$\text{L - CSLP}_1 : \max_{x_1} v_1^1 = 0.3338x_1 + 0.1252x_2 + 0.1252x_3,$$

$$\text{M - CSLP}_1 : \max_{y_1} v_1^2 = 0.6396y_1 + 0.2798y_2 + 0.2998y_3,$$

$$\text{U - CSLP}_1 : \max_{z_1} v_1^3 = 1.2089z_1 + 0.5469z_2 + 0.6045z_3,$$

where $(x_2, y_2, z_2), (x_3, y_3, z_3)$ solves,

subject to

$$x \in G_d,$$

$$v_1^1 \leq v_1^2, v_1^2 \leq v_1^3$$

The individual optimal solutions for the objective functions v_1^{ℓ} ($\ell = 1, 2, 3$) of an FLDM problem are given in Table 2.

Calculating the harmonic mean of the optimum values of the objective functions v_1^{ℓ} ($\ell = 1, 2, 3$), we obtain $HM^F = 2.5443$.

The equivalent single objective function of an FLDM problem using the mean harmonic technique [29] can be formulated as follows:

[FLDM]:

$$\begin{aligned} \text{SLP}_1 : \max_{(x_1, y_1, z_1)} \xi_1 \\ = 0.1312x_1 + 0.0492x_2 + 0.0492x_3 + 0.2514y_1 \\ + 0.1200y_2 + 0.1178y_3 + 0.4751z_1 + 0.2149z_2 \\ + 0.2376z_3, \end{aligned}$$

where $(x_2, y_2, z_2), (x_3, y_3, z_3)$ solves,

subject to

$$x \in G_d,$$

$$\begin{aligned} 0.1312x_1 + 0.0492x_2 + 0.0492x_3 \\ \leq 0.2514y_1 + 0.1200y_2 + 0.1178y_3, \end{aligned}$$

$$\begin{aligned} 0.2514y_1 + 0.1200y_2 + 0.1178y_3 \\ \leq 0.4751z_1 + 0.2149z_2 + 0.2376z_3. \end{aligned}$$

The efficient fuzzy solutions to the FLDM are $\tilde{x}_1^F = (3.5357, 3.5357, 3.7500)$, $\tilde{x}_2^F = (2.1964, 2.5714, 3.2500)$ and $\tilde{x}_3^F = (0, 0, 0)$.

Second, using the FLDM variable $\tilde{x}_1^F = (3.5357, 3.5357, 3.7500)$ in the SLDM, a crisp multiobjective linear programming problem for the SLDM, CMLP₂, can be formulated as follows:

[SLDM]:

$$\begin{aligned} \text{L - CMLP}_2 : \max_{x_2} (u_{21}^1, u_{22}^1) \\ = (12x_1 + 6x_2 + 4x_3, 9x_1 + x_2 + 3x_3), \end{aligned}$$

$$\begin{aligned} \text{M - CMLP}_2 : \max_{y_2} (u_{21}^2, u_{22}^2) \\ = (14y_1 + 7y_2 + 5y_3, 12y_1 + 2y_2 + 6y_3), \end{aligned}$$

$$\begin{aligned} \text{U - CMLP}_2 : \max_{z_2} (u_{21}^3, u_{22}^3) \\ = (17z_1 + 10z_2 + 8z_3, 16z_1 + 4z_2 + 8z_3), \end{aligned}$$

where (x_3, y_3, z_3) solves,

subject to

$$x \in G_d$$

$$x_1 = 3.5357, y_1 = 3.5357, z_1 = 3.7500.$$

However, as obtained in FLDM, the efficient fuzzy solution to an SLDM problem is $\tilde{x}_1^F = (3.5357, 3.5357, 3.75)$, $\tilde{x}_2^S = (2.1964, 2.5714, 3.25)$ and $\tilde{x}_3^S = (0, 0, 0)$.

Table 1 Individual optimal solutions for (x_j, y_j, z_j) and u_t^ℓ of an FLDM problem.

L – CMLP ₁			M – CMLP ₁				U – CMLP ₁				
	u_{11}^1	u_{12}^1	u_{13}^1		u_{11}^2	u_{12}^2	u_{13}^2		u_{11}^3	u_{12}^3	u_{13}^3
x_1	3.5714	3.5714	4.0000	y_1	3.5714	3.5714	4.5000	z_1	3.7500	3.7500	4.5000
x_2	2.1429	2.5714	0.0000	y_2	2.4762	2.4762	0.0000	z_2	3.2500	3.2500	1.0000
x_3	0.0000	0.0000	0.0000	y_3	0.0000	0.0000	0.0000	z_3	0.0000	0.0000	0.0000
Optimal solution	27.8571	39.2857	40.0000		42.0476	56.6190	54.0000		65.0000	78.5000	73.5000

Table 2 Individual optimal solutions for $\mathbf{v}_1^\ell$ of an FLDM problem.

FLDM						
	x_1	3.5714	y_1	3.5714	z_1	3.7500
	x_2	2.1429	y_2	2.4762	z_2	3.2500
	x_3	0.0000	y_3	0.0000	z_3	0.0000
Optimal solution	$\mathbf{v}_1^1$	1.4604	$\mathbf{v}_1^2$	2.9771	$\mathbf{v}_1^3$	6.3108

Third, using the FLDM variable $\tilde{x}_1^F = (3.5357, 3.5357, 3.75)$ and the SLDM variable $\tilde{x}_2^S = (2.1964, 2.5714, 3.25)$ in a TLDM, a crisp multiobjective linear programming problem for the TLDM, CMLP₃, can be formulated as follows: [TLDM]:

$$\begin{aligned} \text{L-CMLP}_3 : \quad & \max_{x_3} (u_{31}^1, u_{32}^1, u_{33}^1) \\ & = (x_1 + 2x_2 + 5x_3, 2x_1 + 3x_2 + 7x_3, 3x_1 + 3x_2 + 10x_3), \end{aligned}$$

$$\begin{aligned} \mathbf{M} - \text{CMLP}_3 : \max_{y_3} & (\mu_{31}^2, u_{32}^2, u_{33}^2) \\ & = (3y_1 + 7y_2 + 14y_3, 8y_1 + 9y_2 + 18y_3, 9y_1 + 11y_2 + 22y_3), \end{aligned}$$

$$\begin{aligned} \text{U-CMLP}_3 : \max_{z_3} (u_{31}^3, u_{32}^3, u_{33}^3) \\ = (6z_1 + 10z_2 + 16z_3, 10z_1 + 12z_2 + 20z_3, 11z_1 + 13z_2 + 25z_3), \end{aligned}$$

subject to

$$x \in G_d,$$

$$x_1 = 3.5357, y_1 = 3.5357, z_1 = 3.7500, x_2 = 2.1964, \\ y_2 = 2.5714, z_2 = 3.2500.$$

The TLDM can be treated similarly, as approached in the FLDM. Then, the TLDM efficient fuzzy solution is $\tilde{x}_1^F = (3.5357, 3.5357, 3.7500)$, $\tilde{x}_2^S = (2.1964, 2.5714, 3.2500)$ and $\tilde{x}_3^T = (0, 0, 0)$.

Finally, the fuzzy compromise solution for the fully fuzzy model given in the illustrated example is obtained as follows: $\tilde{x}_* = (3.5357, 3.5357, 3.7500)$.

$$\tilde{x}_2^* = (2.1964, 2.5714, 3.2500), \quad \tilde{x}_3^* = (0, 0, 0),$$

$$\tilde{f}_{11}^* = (27.8034, 42.1069, 65), \quad \tilde{f}_{12}^* = (39.2676, 56.8925, 78.5),$$

$$f_{13} = (37.5534, 50.1426, 75.75),$$

$$\tilde{f}_{21}^* = (55.6068, 67.4996, 96.25), \quad \tilde{f}_{22}^* = (34.0177, 47.5712, 73),$$

$$\begin{aligned} \tilde{f}_{31}^* &= (7.4641, 28.6069, 55), & \tilde{f}_{32}^* &= (12.9640, 51.4282, 76.5), \\ \tilde{f}_{33}^* &= (16.4997, 60.1067, 83.5). \end{aligned}$$

5.2. Application example on network product supply chain

Consider Lee et al.'s [18] example on a three-level and five-objective network product supply chain in a fully fuzzy environment to explain the application of the suggested FFMMLP model and solution procedure. The related data of this example are given in Tables 3–5.

The objective function of the wholesaler in crisp form is formulated as follows:

$$\begin{aligned} \mathbf{L} - \text{CMLP}_1 : \max Z_1^* = & 9224700 \\ & - (510Q_{11}^* + 390Q_{21}^* + 510Q_{12}^* + 390Q_{22}^* + 510Q_{13}^* + 390Q_{23}^* + 510Q_{14}^* \\ & + 390Q_{24}^* + 80H_{11}^{(w)} + 50H_{21}^{(w)} + 80H_{12}^{(w)} + 50H_{22}^{(w)} + 80H_{13}^{(w)} + 50H_{23}^{(w)} \\ & + 80H_{14}^{(w)} + 50H_{24}^{(w)}) \end{aligned}$$

$$\begin{aligned} \mathbf{M} - \text{CMLP}_1 : \max Z_1^* &= 12460400 \\ &- (440Q_{11}^{(w)} + 360Q_{21}^{(w)} + 440Q_{12}^{(w)} + 360Q_{22}^{(w)} + 440Q_{13}^{(w)} + 360Q_{23}^{(w)} \\ &+ 440Q_{14}^{(w)} + 360Q_{24}^{(w)} + 70H_{11}^{2(w)} + 30F_{21}^{2(w)} + 70H_{12}^{2(w)} + 30H_{22}^{2(w)} + 70H_{13}^{2(w)} \\ &+ 30H_{23}^{2(w)} + 70H_{14}^{2(w)} + 30H_{24}^{2(w)}) \end{aligned}$$

$$\begin{aligned} & \mathbf{U} - \text{CMLP}_1 : \max \mathbf{Z}_1^3 = 14700100 \\ & - (370\mathcal{Q}_{11}^1 + 330\mathcal{Q}_{21}^1 + 370\mathcal{Q}_{12}^1 + 330\mathcal{Q}_{22}^1 + 370\mathcal{Q}_{13}^1 + 330\mathcal{Q}_{23}^1 + 370\mathcal{Q}_{14}^1 \\ & + 330\mathcal{Q}_{24}^1 + 60H_{11}^{(w)} + 15H_{21}^{(w)} + 60H_{12}^{(w)} + 15H_{22}^{(w)} + 60H_{13}^{(w)} + 15H_{23}^{(w)} \\ & + 60H_{14}^{(w)} + 15H_{24}^{(w)}), \end{aligned}$$

subject to

$$H_{11}^{1(w)} = 100 + Q_{11}^1 - 1800, H_{11}^{2(w)} = 100 + Q_{11}^2 - 1650,$$

$$H_{21}^{1(w)} = 100 + Q_{21}^1 - 940, \quad H_{21}^{2(w)} = 100 + Q_{21}^2 - 870,$$

Table 3 Data of the wholesaler.

Period	Product	$\tilde{p}_{it}^{(w)}$	$\tilde{Q}_{it}^{(w)}$	$\tilde{P}_{it}$	$\tilde{h}_{it}^{(w)}$	$\tilde{\varepsilon}_{it}^{(w)}$	$\tilde{H}_{it \max}^{(w)}$
1	1	(750, 870, 920)	(1500, 1650, 1800)	(370, 440, 510)	(60, 70, 80)	(65, 80, 95)	(600, 600, 600)
	2	(600, 730, 810)	(700, 870, 940)	(330, 360, 390)	(15, 30, 50)	(65, 80, 95)	(500, 500, 500)
2	1	(750, 870, 920)	(1920, 2010, 2080)	(370, 440, 510)	(60, 70, 80)	(65, 80, 95)	(600, 600, 600)
	2	(600, 730, 810)	(1470, 1560, 1660)	(330, 360, 390)	(15, 30, 50)	(65, 80, 95)	(500, 500, 500)
3	1	(750, 870, 920)	(4440, 4600, 4890)	(370, 440, 510)	(60, 70, 80)	(65, 80, 95)	(600, 600, 600)
	2	(600, 730, 810)	(1630, 1790, 1920)	(330, 360, 390)	(15, 30, 50)	(65, 80, 95)	(500, 500, 500)
4	1	(750, 870, 920)	(1800, 2000, 2100)	(370, 440, 510)	(60, 70, 80)	(65, 80, 95)	(600, 600, 600)
	2	(600, 730, 810)	(2340, 2480, 2550)	(330, 360, 390)	(15, 30, 50)	(65, 80, 95)	(500, 500, 500)

$\tilde{H}_{10}^{(w)} = (100, 100, 100)$, $\tilde{H}_{20}^{(w)} = (100, 100, 100)$.

Table 4 Data of the manufacture.

Period	Product	$\tilde{p}_{it}$	$\tilde{\phi}_{it}$	$\tilde{s}_{it}$	$\tilde{\mathcal{T}}_{it}$	$\tilde{h}_{it}$	$\tilde{l}_{it}$
1	1	(300, 450, 520)	(90, 150, 200)	(200, 250, 300)	(200, 230, 260)	(100, 100, 100)	(0.1, 0.3, 0.5)
	2	(280, 350, 430)	(50, 70, 90)	(90, 120, 150)	(80, 100, 120)	(100, 100, 100)	(0.2, 0.4, 0.6)
2	1	(300, 450, 520)	(90, 150, 200)	(200, 250, 300)	(200, 230, 260)	(100, 100, 100)	(0.1, 0.3, 0.5)
	2	(280, 350, 430)	(50, 70, 90)	(90, 120, 150)	(80, 100, 120)	(100, 100, 100)	(0.2, 0.4, 0.6)
3	1	(300, 450, 520)	(90, 150, 200)	(200, 250, 300)	(200, 230, 260)	(100, 100, 100)	(0.1, 0.3, 0.5)
	2	(280, 350, 430)	(50, 70, 90)	(90, 120, 150)	(80, 100, 120)	(100, 100, 100)	(0.2, 0.4, 0.6)
4	1	(300, 450, 520)	(90, 150, 200)	(200, 250, 300)	(200, 230, 260)	(100, 100, 100)	(0.1, 0.3, 0.5)
	2	(280, 350, 430)	(50, 70, 90)	(90, 120, 150)	(80, 100, 120)	(100, 100, 100)	(0.2, 0.4, 0.6)
Period	Product	$\tilde{L}_{it \max}$	$\tilde{E}_{it \max}$	$\tilde{W}_{it \max}$	$\tilde{e}_{it}$	$\tilde{w}_i$	$\tilde{w}_j$
1	1	(100, 200, 300)	(480, 550, 600)	(8000, 9000, 10000)	(0.02, 0.05, 0.1)	(1, 3, 5)	(2, 2, 2)
	2				(0.02, 0.05, 0.08)	(1, 3, 5)	(2, 2, 2)
2	1	(100, 200, 300)	(450, 500, 580)	(8000, 9000, 10000)	(0.02, 0.05, 0.1)	(1, 3, 5)	(2, 2, 2)
	2				(0.02, 0.05, 0.08)	(1, 3, 5)	(2, 2, 2)
3	1	(100, 200, 300)	(500, 550, 610)	(8000, 9000, 10000)	(0.02, 0.05, 0.1)	(1, 3, 5)	(2, 2, 2)
	2				(0.02, 0.05, 0.08)	(1, 3, 5)	(2, 2, 2)
4	1	(100, 200, 300)	(450, 500, 580)	(8000, 9000, 10000)	(0.02, 0.05, 0.1)	(1, 3, 5)	(2, 2, 2)
	2				(0.02, 0.05, 0.08)	(1, 3, 5)	(2, 2, 2)

$\tilde{H}_{10} = (0, 0, 0)$, $\tilde{H}_{20} = (0, 0, 0)$, $\tilde{H}_{10}^{(m)} = (800, 900, 1000)$, $\tilde{H}_{20}^{(m)} = (800, 900, 1000)$, $\tilde{H}_{30}^{(m)} = (800, 900, 1000)$
The minimum inventory level of product 1 or product 2 is (100, 100, 100) units in each period.

$$\begin{aligned}
H_{12}^{1(w)} &= H_{11}^{1(w)} + Q_{12}^1 - 2080, & H_{12}^{2(w)} &= H_{11}^{2(w)} + Q_{12}^2 - 2010, & H_{2t}^{1(w)} &\leq 500, (t = 1, 2, 3, 4), & H_{2t}^{2(w)} &\leq 500, (t = 1, 2, 3, 4), \\
H_{22}^{1(w)} &= H_{21}^{1(w)} + Q_{22}^1 - 1660, & H_{22}^{2(w)} &= H_{21}^{2(w)} + Q_{22}^2 - 1560, & H_{11}^{3(w)} &= 100 + Q_{11}^3 - 1500, \\
H_{13}^{1(w)} &= H_{12}^{1(w)} + Q_{13}^1 - 4890, & H_{13}^{2(w)} &= H_{12}^{2(w)} + Q_{13}^2 - 4600, & H_{21}^{3(w)} &= 100 + Q_{21}^3 - 700, \\
H_{23}^{1(w)} &= H_{22}^{1(w)} + Q_{23}^1 - 1920, & H_{23}^{2(w)} &= H_{22}^{2(w)} + Q_{23}^2 - 1790, & H_{12}^{3(w)} &= H_{11}^{3(w)} + Q_{12}^3 - 1920, \\
H_{14}^{1(w)} &= H_{13}^{1(w)} + Q_{14}^1 - 2100, & H_{14}^{2(w)} &= H_{13}^{2(w)} + Q_{14}^2 - 2000, & H_{22}^{3(w)} &= H_{21}^{3(w)} + Q_{22}^3 - 1470, \\
H_{24}^{1(w)} &= H_{23}^{1(w)} + Q_{24}^1 - 2550, & H_{24}^{2(w)} &= H_{23}^{2(w)} + Q_{24}^2 - 2480, & H_{13}^{3(w)} &= H_{12}^{3(w)} + Q_{13}^3 - 4440, \\
H_{1t}^{1(w)} &\leq 600, (t = 1, 2, 3, 4), & H_{1t}^{2(w)} &\leq 600, (t = 1, 2, 3, 4), & H_{23}^{3(w)} &= H_{22}^{3(w)} + Q_{23}^3 - 1630, \\
&&&&&&&&&& H_{14}^{3(w)} &= H_{13}^{3(w)} + Q_{14}^3 - 1800, \\
&&&&&&&&&& H_{24}^{3(w)} &= H_{23}^{3(w)} + Q_{24}^3 - 2340,
\end{aligned}$$

Table 5 Data of the supplier first period.

$\tilde{p}_{1t}^{(s)}$	$\tilde{p}_{2t}^{(s)}$	$\tilde{p}_{3t}^{(s)}$	$\tilde{\theta}_{1t}^{(s)}$	$\tilde{\theta}_{2t}^{(s)}$	$\tilde{\theta}_{3t}^{(s)}$	$\tilde{h}_{1t}^{(s)}$	$\tilde{h}_{2t}^{(s)}$
(40, 60, 80)	(50, 65, 80)	(60, 70, 90)	(1, 2, 4)	(1, 3, 5)	(1, 2, 3)	(1, 2, 3)	(1, 1, 1)
$\tilde{h}_{3t}^{(s)}$	$\tilde{H}_{1tmax}^{(s)}$	$\tilde{H}_{2tmax}^{(s)}$	$\tilde{H}_{3tmax}^{(s)}$	$\tilde{\varepsilon}_{1t}^{(s)}$	$\tilde{\varepsilon}_{2t}^{(s)}$	$\tilde{\varepsilon}_{3t}^{(s)}$	
(1, 2, 4)	(8000, 9000, 10000)	(8000, 9000, 10000)	(8000, 9000, 10000)	(1, 2, 3)	(1, 2, 3)	(1, 2, 3)	
$\tilde{H}_{10} = (1000, 1500, 2000)$, $\tilde{H}_{20} = (1000, 1500, 2000)$, $\tilde{H}_{30} = (1000, 1500, 2000)$, $\tilde{H}_{14} = (1000, 2000, 3000)$, $\tilde{H}_{24} = (1000, 2000, 3000)$, $\tilde{H}_{34} = (1000, 2000, 3000)$. Data of the other three periods are the same as those of the first period.							

Table 6 The efficient fuzzy solutions of the wholesaler.

ℓ	Q'_{11}	Q'_{21}	Q'_{12}	Q'_{22}	Q'_{13}	Q'_{23}	Q'_{14}	Q'_{24}	H'_{11}	H'_{21}	H'_{12}	H'_{22}	H'_{13}	H'_{23}	H'_{14}	H'_{24}	Z'_1
1	1700	840	2080	1660	4890	1920	2100	2550	0	0	0	0	0	0	0	0	1,993,500
2	1550	770	2010	1560	4600	1790	2000	2480	0	0	0	0	0	0	0	0	5,614,000
3	1400	600	1920	1470	4440	1630	1800	2340	0	0	0	0	0	0	0	0	8,415,100

$$H_{1t}^{3(w)} \leq 600, (t = 1, 2, 3, 4)$$

$$H_{2t}^{3(w)} \leq 500, (t = 1, 2, 3, 4).$$

In Table 6 shows the efficient fuzzy solutions of the wholesaler.

Using the efficient fuzzy solutions of the wholesaler in the manufacturer, the objective functions of manufacturer in crisp form are formulated as follows:

L – CMLP₂ :

$$\begin{aligned} \max Z_2^1 = & 5182600 - 200X_{11}^3 - 90X_{21}^3 - 200X_{12}^3 - 90X_{22}^3 \\ & - 200X_{13}^3 - 90X_{23}^3 - 200X_{14}^3 - 90X_{24}^3 - 260Y_{11}^3 \\ & - 120Y_{21}^3 - 260Y_{12}^3 - 120Y_{22}^3 - 260Y_{13}^3 - 120Y_{23}^3 \\ & - 260Y_{14}^3 - 120Y_{24}^3 - 300S_{11}^3 - 150S_{21}^3 - 300S_{12}^3 \\ & - 150S_{22}^3 - 300S_{13}^3 - 150S_{23}^3 - 300S_{14}^3 - 150S_{24}^3 \\ & - 100H_{11}^3 - 100H_{21}^3 - 100H_{12}^3 - 100H_{22}^3 - 100H_{13}^3 \\ & - 100H_{23}^3 - 100H_{14}^3 - 100H_{24}^3, \end{aligned}$$

$$\begin{aligned} \min Z_3^1 = & H_{11}^1 + H_{21}^1 + H_{12}^1 + H_{22}^1 + H_{13}^1 + H_{23}^1 + H_{14}^1 + H_{24}^1 \\ & + H_{11}^{1(m)} + H_{12}^{1(m)} + H_{13}^{1(m)} + H_{14}^{1(m)} + H_{21}^{1(m)} + H_{22}^{1(m)} \\ & + H_{23}^{1(m)} + H_{24}^{1(m)} + H_{31}^{1(m)} + H_{32}^{1(m)} + H_{33}^{1(m)} + H_{34}^{1(m)}, \end{aligned}$$

$$\min Z_4^1 = A_2^1 + A_3^1 + A_4^1 + B_2^1 + B_3^1 + B_4^1,$$

M – CMLP₂ :

$$\begin{aligned} \max Z_2^2 = & 6882000 - 150X_{11}^2 - 70X_{21}^2 - 150X_{12}^2 - 70X_{22}^2 \\ & - 150X_{13}^2 - 70X_{23}^2 - 150X_{14}^2 - 70X_{24}^2 - 230Y_{11}^2 \\ & - 100Y_{21}^2 - 230Y_{12}^2 - 100Y_{22}^2 - 230Y_{13}^2 - 100Y_{23}^2 \\ & - 230Y_{14}^2 - 100Y_{24}^2 - 250S_{11}^2 - 120S_{21}^2 - 250S_{12}^2 \\ & - 120S_{22}^2 - 250S_{13}^2 - 120S_{23}^2 - 250S_{14}^2 - 120S_{24}^2 \\ & - 100H_{11}^2 - 100H_{21}^2 - 100H_{12}^2 - 100H_{22}^2 - 100H_{13}^2 \\ & - 100H_{23}^2 - 100H_{14}^2 - 100H_{24}^2, \end{aligned}$$

$$\begin{aligned} \min Z_3^2 = & H_{11}^2 + H_{21}^2 + H_{12}^2 + H_{22}^2 + H_{13}^2 + H_{23}^2 + H_{14}^2 + H_{24}^2 \\ & + H_{11}^{2(m)} + H_{12}^{2(m)} + H_{13}^{2(m)} + H_{14}^{2(m)} + H_{21}^{2(m)} + H_{22}^{2(m)} \\ & + H_{23}^{2(m)} + H_{24}^{2(m)} + H_{31}^{2(m)} + H_{32}^{2(m)} + H_{33}^{2(m)} + H_{34}^{2(m)}, \end{aligned}$$

$$\min Z_4^2 = A_2^2 + A_3^2 + A_4^2 + B_2^2 + B_3^2 + B_4^2,$$

U – CMLP₂ :

$$\begin{aligned} \max Z_2^3 = & 7568400 - 90X_{11}^1 - 50X_{21}^1 - 90X_{12}^1 - 50X_{22}^1 - 90X_{13}^1 \\ & - 50X_{23}^1 - 90X_{14}^1 - 50X_{24}^1 - 200Y_{11}^1 - 80Y_{21}^1 \\ & - 200Y_{12}^1 - 80Y_{22}^1 - 200Y_{13}^1 - 80Y_{23}^1 - 200Y_{14}^1 \\ & - 80Y_{24}^1 - 200S_{11}^1 - 90S_{21}^1 - 200S_{12}^1 - 90S_{22}^1 \\ & - 200S_{13}^1 - 90S_{23}^1 - 200S_{14}^1 - 90S_{24}^1 - 100H_{11}^1 \\ & - 100H_{21}^1 - 100H_{12}^1 - 100H_{22}^1 - 100H_{13}^1 - 100H_{23}^1 \\ & - 100H_{14}^1 - 100H_{24}^1, \end{aligned}$$

$$\begin{aligned} \min Z_3^3 = & H_{11}^3 + H_{21}^3 + H_{12}^3 + H_{22}^3 + H_{13}^3 + H_{23}^3 + H_{14}^3 + H_{24}^3 \\ & + H_{11}^{3(m)} + H_{12}^{3(m)} + H_{13}^{3(m)} + H_{14}^{3(m)} + H_{21}^{3(m)} + H_{22}^{3(m)} \\ & + H_{23}^{3(m)} + H_{24}^{3(m)} + H_{31}^{3(m)} + H_{32}^{3(m)} + H_{33}^{3(m)} + H_{34}^{3(m)}, \end{aligned}$$

$$\min Z_4^3 = A_2^3 + A_3^3 + A_4^3 + B_2^3 + B_3^3 + B_4^3,$$

Subject to

$$H_{11}^1 = X_{11}^1 + Y_{11}^1 + S_{11}^1 - Q_{11}^1,$$

$$H_{21}^1 = X_{21}^1 + Y_{21}^1 + S_{21}^1 - Q_{21}^1,$$

$$H_{it}^1 = H_{i(t-1)}^1 + X_{it}^1 + Y_{it}^1 + S_{it}^1 - Q_{it}^1, (i = 1, 2; t = 2, 3, 4)$$

$$H_{it}^1 \geq 100, (i = 1, 2; t = 1, 2, 3, 4)$$

$$0.1(X_{1t}^1 + Y_{1t}^1) + 0.2(X_{2t}^1 + Y_{2t}^1) \leq 800, (t = 1, 2, 3, 4)$$

$$0.02(X_{11}^1 + Y_{11}^1) + 0.02(X_{21}^1 + Y_{21}^1) \leq 480$$

$$0.02(X_{12}^1 + Y_{12}^1) + 0.02(X_{22}^1 + Y_{22}^1) \leq 450$$

$$0.02(X_{13}^1 + Y_{13}^1) + 0.02(X_{23}^1 + Y_{23}^1) \leq 500$$

$$0.02(X_{14}^1 + Y_{14}^1) + 0.02(X_{24}^1 + Y_{24}^1) \leq 450$$

$$H_{1t}^1 + H_{2t}^1 + 2H_{1t}^{1(m)} + 2H_{2t}^{1(m)} + 2H_{3t}^{1(m)} \leq 8000, \quad (t = 1, 2, 3, 4)$$

$$Q_{j1}^1 = 800 + 3(X_{11}^1 + Y_{11}^1 + X_{21}^1 + Y_{21}^1) - H_{j1}^{3(m)}, \quad (j = 1, 2, 3)$$

$$Q_{jt}^1 = H_{j(t-1)}^{1(m)} + 3(X_{1t}^1 + Y_{1t}^1 + X_{2t}^1 + Y_{2t}^1) - H_{jt}^{3(m)}, \quad (j = 1, 2, 3; t = 2, 3, 4)$$

$$0.1(X_{1(t-1)}^1 + Y_{1(t-1)}^1) + 0.2(X_{2(t-1)}^1 + Y_{2(t-1)}^1) + 8A_t^1 - 8B_t^3 \\ = 0.1(X_{1t}^1 + Y_{1t}^1) + 0.2(X_{2t}^1 + Y_{2t}^1), \quad (t = 1, 2, 3, 4)$$

$$H_{11}^2 = X_{11}^2 + Y_{11}^2 + S_{11}^2 - Q_{11}^2,$$

$$H_{11}^2 = X_{11}^2 + Y_{11}^2 + S_{11}^2 - Q_{11}^2,$$

$$H_{it}^2 = H_{i(t-1)}^2 + X_{it}^2 + Y_{it}^2 + S_{it}^2 - Q_{it}^2, \quad (i = 1, 2; t = 2, 3, 4)$$

$$H_{it}^2 \geq 100, \quad (i = 1, 2; t = 1, 2, 3, 4)$$

$$0.3(X_{1t}^2 + Y_{1t}^2) + 0.4(X_{2t}^2 + Y_{2t}^2) \leq 1600, \quad (t = 1, 2, 3, 4)$$

$$0.05(X_{11}^2 + Y_{11}^2) + 0.05(X_{21}^2 + Y_{21}^2) \leq 550$$

$$0.05(X_{12}^2 + Y_{12}^2) + 0.05(X_{22}^2 + Y_{22}^2) \leq 500$$

$$0.05(X_{13}^2 + Y_{13}^2) + 0.05(X_{23}^2 + Y_{23}^2) \leq 550$$

$$0.05(X_{14}^2 + Y_{14}^2) + 0.05(X_{24}^2 + Y_{24}^2) \leq 500$$

$$3H_{1t}^2 + 3H_{2t}^2 + 2H_{1t}^{2(m)} + 2H_{2t}^{2(m)} + 2H_{3t}^{2(m)} \\ \leq 9000, \quad (t = 1, 2, 3, 4)$$

$$Q_{j1}^2 = 900 + 3(X_{11}^2 + Y_{11}^2 + X_{21}^2 + Y_{21}^2) - H_{j1}^{2(m)}, \quad (j = 1, 2, 3)$$

$$Q_{jt}^2 = H_{j(t-1)}^{2(m)} + 3(X_{1t}^2 + Y_{1t}^2 + X_{2t}^2 + Y_{2t}^2) - H_{jt}^{2(m)}, \quad (j = 1, 2, 3; t = 2, 3, 4)$$

$$0.3(X_{1(t-1)}^2 + Y_{1(t-1)}^2) + 0.4(X_{2(t-1)}^2 + Y_{2(t-1)}^2) + 8A_t^2 - 8B_t^2 \\ = 0.3(X_{1t}^2 + Y_{1t}^2) + 0.4(X_{2t}^2 + Y_{2t}^2), \quad (t = 1, 2, 3, 4)$$

$$H_{11}^3 = X_{11}^3 + Y_{11}^3 + S_{11}^3 - Q_{11}^1,$$

$$H_{21}^3 = X_{21}^3 + Y_{21}^3 + S_{21}^3 - Q_{21}^1,$$

$$H_{it}^3 = H_{i(t-1)}^3 + X_{it}^3 + Y_{it}^3 + S_{it}^3 - Q_{it}^1, \quad (i = 1, 2; t = 2, 3, 4)$$

$$H_{it}^3 \geq 100, \quad (i = 1, 2; t = 1, 2, 3, 4)$$

$$0.5(X_{1t}^3 + Y_{1t}^3) + 0.6(X_{2t}^3 + Y_{2t}^3) \leq 2400, \quad (t = 1, 2, 3, 4)$$

$$0.1(X_{11}^3 + Y_{11}^3) + 0.08(X_{21}^3 + Y_{21}^3) \leq 600$$

$$0.1(X_{12}^3 + Y_{12}^3) + 0.08(X_{22}^3 + Y_{22}^3) \leq 580$$

$$0.1(X_{13}^3 + Y_{13}^3) + 0.08(X_{23}^3 + Y_{23}^3) \leq 610$$

$$0.1(X_{14}^3 + Y_{14}^3) + 0.08(X_{24}^3 + Y_{24}^3) \leq 580$$

$$5H_{1t}^3 + 5H_{2t}^3 + 2H_{1t}^{3(m)} + 2H_{2t}^{3(m)} + 2H_{3t}^{3(m)} \\ \leq 10000, \quad (t = 1, 2, 3, 4)$$

$$Q_{j1}^3 = 1000 + 3(X_{11}^3 + Y_{11}^3 + X_{21}^3 + Y_{21}^3) - H_{j1}^{1(m)}, \quad (j = 1, 2, 3)$$

$$Q_{jt}^3 = H_{j(t-1)}^{3(m)} + 3(X_{1t}^3 + Y_{1t}^3 + X_{2t}^3 + Y_{2t}^3) - H_{jt}^{1(m)}, \quad (j = 1, 2, 3; t = 2, 3, 4)$$

$$0.5(X_{1(t-1)}^3 + Y_{1(t-1)}^3) + 0.6(X_{2(t-1)}^3 + Y_{2(t-1)}^3) + 8A_t^3 - 8B_t^1 \\ = 0.5(X_{1t}^3 + Y_{1t}^3) + 0.6(X_{2t}^3 + Y_{2t}^3), \quad (t = 1, 2, 3, 4).$$

Table 7 shows the efficient fuzzy solutions of the manufacturer.

Using the efficient fuzzy solutions of the manufacturer in the supplier, the objective functions of the supplier in crisp form are formulated as follows:

$$\begin{aligned} \text{L - CMLP}_3 : \max Z_5^1 \\ = 67466.6667 - 4(Q_{11}^{3(s)} + Q_{12}^{3(s)} + Q_{13}^{3(s)} + Q_{14}^{3(s)}) \\ - 5(Q_{21}^{3(s)} + Q_{22}^{3(s)} + Q_{23}^{3(s)} + Q_{24}^{3(s)}) \\ - 3(Q_{31}^{3(s)} + Q_{32}^{3(s)} + Q_{33}^{3(s)} + Q_{34}^{3(s)}) \\ - 3(H_{11}^{3(s)} + H_{12}^{3(s)} + H_{13}^{3(s)} + H_{14}^{3(s)}) \\ - (H_{21}^{3(s)} + H_{22}^{3(s)} + H_{23}^{3(s)} + H_{24}^{3(s)}) \\ - 4(H_{31}^{3(s)} + H_{32}^{3(s)} + H_{33}^{3(s)} + H_{34}^{3(s)}) \end{aligned}$$

$$\begin{aligned} \text{M - CMLP}_3 : \max Z_5^2 \\ = 102000 - 2(Q_{11}^{2(s)} + Q_{12}^{2(s)} + Q_{13}^{2(s)} + Q_{14}^{2(s)}) \\ - 3(Q_{21}^{2(s)} + Q_{22}^{2(s)} + Q_{23}^{2(s)} + Q_{24}^{2(s)}) \\ - 2(Q_{31}^{2(s)} + Q_{32}^{2(s)} + Q_{33}^{2(s)} + Q_{34}^{2(s)}) \\ - 2(H_{11}^{2(s)} + H_{12}^{2(s)} + H_{13}^{2(s)} + H_{14}^{2(s)}) \\ - (H_{21}^{2(s)} + H_{22}^{2(s)} + H_{23}^{2(s)} + H_{24}^{2(s)}) \\ - 2(H_{31}^{2(s)} + H_{32}^{2(s)} + H_{33}^{2(s)} + H_{34}^{2(s)}) \end{aligned}$$

Table 7 The efficient fuzzy solutions of the manufacturer.

ℓ	X'_{11}	X'_{21}	X'_{12}	X'_{22}	X'_{13}	X'_{23}	X'_{14}	X'_{24}	Y'_{11}	Y'_{21}	Y'_{12}	Y'_{22}	Y'_{13}	Z'_2	
1	100	100	0	0	0	0	0	0	0	0	0	0	0	5,064,600	
2	100	100	0	0	0	0	0	0	0	0	0	0	0	6,882,000	
3	100	0	0	0	0	0	0	33.33	0	0	0	0	0	7,568,400	
ℓ	Y'_{23}	Y'_{14}	Y'_{24}	S'_{11}	S'_{21}	S'_{12}	S'_{22}	S'_{13}	S'_{23}	S'_{14}	S'_{24}	H'_{11}	H'_{21}	Z'_3	
1	0	0	0	0	0	0	0	0	0	0	0	100	100	4200	
2	0	0	0	0	0	0	0	0	0	0	0	100	100	4400	
3	0	0	0	0	100	0	0	0	0	0	0	100	100	4600	
ℓ	H'_{12}	H'_{22}	H'_{13}	H'_{23}	H'_{14}	H'_{24}	$H^{(m)}_{11}$	$H^{(m)}_{12}$	$H^{(m)}_{13}$	$H^{(m)}_{14}$	$H^{(m)}_{21}$	$H^{(m)}_{22}$	$H^{(m)}_{23}$	Z'_4	
1	100	100	100	100	100	100	1300	100	100	200	1300	100	100	0.875	
2	100	100	100	100	100	100	1500	100	100	100	1500	100	100	0.875	
3	100	100	100	100	100	100	1400	100	100	100	1400	100	100	0.875	
ℓ	$H^{(m)}_{24}$	$H^{(m)}_{31}$	$H^{(m)}_{32}$	$H^{(m)}_{33}$	$H^{(m)}_{34}$	X'_{10}	X'_{20}	Y'_{10}	Y'_{20}	Q'_{11}	Q'_{12}	Q'_{13}	Q'_{14}	Q'_{21}	
1	200	0	0	0	0	0	0	0	0	0	0	0	0	0	
2	100	0	0	0	0	0	0	0	0	0	0	0	0	0	
3	66.67	233.3	0	0	0	0	0	0	0	0	0	0	0	0	
ℓ	Q'_{22}	Q'_{23}	Q'_{24}	Q'_{31}	Q'_{32}	Q'_{33}	Q'_{34}	B'_1	B'_2	B'_3	B'_4	A'_1	A'_2	A'_3	A'_4
1	0	0	33.33	1166.7	0	0	0	0	0.625	0	0	0.375	0	0	0.25
2	0	0	0	1500	0	0	0	0	0.875	0	0	0.875	0	0	0
3	0	0	0	1300	0	0	100	0	0.375	0	0.25	0.625	0	0	0.25

U – CMLP₃ : $\max Z'_5$

$$\begin{aligned}
&= 124800 - \left(Q^{1(s)}_{11} + Q^{1(s)}_{12} + Q^{1(s)}_{13} + Q^{1(s)}_{14} \right) \\
&\quad - \left(Q^{1(s)}_{21} + Q^{1(s)}_{22} + Q^{1(s)}_{23} + Q^{1(s)}_{24} \right) \\
&\quad - \left(Q^{1(s)}_{31} + Q^{1(s)}_{32} + Q^{1(s)}_{33} + Q^{1(s)}_{34} \right) \\
&\quad - \left(H^{1(s)}_{11} + H^{1(s)}_{12} + H^{1(s)}_{13} + H^{1(s)}_{14} \right) \\
&\quad - \left(H^{1(s)}_{21} + H^{1(s)}_{22} + H^{1(s)}_{23} + H^{1(s)}_{24} \right) \\
&\quad - \left(H^{1(s)}_{31} + H^{1(s)}_{32} + H^{1(s)}_{33} + H^{1(s)}_{34} \right)
\end{aligned}$$

subject to

In Table 8 shows the efficient fuzzy solutions of the supplier.

$$1000 + Q^{1(s)}_{11} = H^{1(s)}_{11}, + Q^{2(s)}_{11} = H^{2(s)}_{11},$$

$$H^{1(s)}_{11} + Q^{1(s)}_{12} = H^{1(s)}_{12}, \quad H^{2(s)}_{11} + Q^{2(s)}_{12} = H^{2(s)}_{12},$$

$$H^{1(s)}_{12} + Q^{1(s)}_{13} = H^{1(s)}_{13}, \quad H^{2(s)}_{12} + Q^{2(s)}_{13} = H^{2(s)}_{13},$$

$$H^{1(s)}_{13} + Q^{1(s)}_{14} = 1000, \quad H^{2(s)}_{13} + Q^{2(s)}_{14} = 2000,$$

$$1000 + Q^{1(s)}_{21} = H^{1(s)}_{21}, \quad 1500 + Q^{2(s)}_{21} = H^{2(s)}_{21},$$

$$H^{1(s)}_{21} + Q^{1(s)}_{22} = H^{1(s)}_{22}, \quad H^{2(s)}_{21} + Q^{2(s)}_{22} = H^{2(s)}_{22},$$

$$H^{1(s)}_{22} + Q^{1(s)}_{23} = H^{1(s)}_{23}, \quad H^{2(s)}_{22} + Q^{2(s)}_{23} = H^{2(s)}_{23},$$

$$H^{1(s)}_{23} + Q^{1(s)}_{24} = 1000, \quad H^{2(s)}_{23} + Q^{2(s)}_{24} = 2000,$$

Table 8 The efficient fuzzy solutions of the supplier.

ℓ	$H^{(s)}_{11}$	$H^{(s)}_{12}$	$H^{(s)}_{13}$	$H^{(s)}_{14}$	$H^{(s)}_{21}$	$H^{(s)}_{22}$	$H^{(s)}_{23}$	$H^{(s)}_{24}$	$H^{(s)}_{31}$	$H^{(s)}_{32}$	$H^{(s)}_{33}$	$H^{(s)}_{34}$	Z'_5
1	2000	2000	2000	0	2000	2000	2000	0	833.33	833.33	833.33	0	17,800
2	1500	1500	1500	0	1500	1500	1500	0	0	0	0	0	82,000
3	1000	1000	1000	0	1000	1000	1000	0	0	0	0	0	117,400
ℓ	$Q^{(s)}_{11}$	$Q^{(s)}_{12}$	$Q^{(s)}_{13}$	$Q^{(s)}_{14}$	$Q^{(s)}_{21}$	$Q^{(s)}_{22}$	$Q^{(s)}_{23}$	$Q^{(s)}_{24}$	$Q^{(s)}_{31}$	$Q^{(s)}_{32}$	$Q^{(s)}_{33}$	$Q^{(s)}_{34}$	
1	0	0	0	1000	0	0	0	1033.33	0	0	0	2166.67	
2	0	0	0	500	0	0	0	500	0	0	0	2000	
3	0	0	0	0	0	0	0	0	300	0	0	1100	

$$\begin{aligned}
-300 + Q_{31}^{1(s)} &= H_{31}^{1(s)}, & Q_{31}^{2(s)} &= H_{31}^{2(s)}, \\
H_{31}^{1(s)} + Q_{32}^{1(s)} &= H_{32}^{1(s)}, & H_{31}^{2(s)} + Q_{32}^{2(s)} &= H_{32}^{2(s)}, \\
H_{32}^{1(s)} + Q_{33}^{1(s)} &= H_{33}^{1(s)}, & H_{32}^{2(s)} + Q_{33}^{2(s)} &= H_{33}^{2(s)}, \\
H_{33}^{1(s)} + Q_{34}^{1(s)} &= 1100, & H_{33}^{2(s)} + Q_{34}^{2(s)} &= 2000, \\
H_{jt}^{1(s)} &\leq 8000, \quad (j = 1, 2, 3; t = 1, 2, 3, 4), \\
H_{jt}^{2(s)} &\leq 9000, \quad (j = 1, 2, 3; t = 1, 2, 3, 4), \\
2000 + Q_{11}^{3(s)} &= H_{11}^{3(s)}, \\
H_{11}^{3(s)} + Q_{12}^{3(s)} &= H_{12}^{3(s)}, \\
H_{12}^{3(s)} + Q_{13}^{3(s)} &= H_{13}^{3(s)}, \\
H_{13}^{3(s)} + Q_{14}^{3(s)} &= 3000, \\
2000 + Q_{21}^{3(s)} &= H_{21}^{3(s)}, \\
H_{21}^{3(s)} + Q_{22}^{3(s)} &= H_{22}^{3(s)}, \\
H_{22}^{3(s)} + Q_{23}^{3(s)} &= H_{23}^{3(s)}, \\
H_{23}^{3(s)} + Q_{24}^{3(s)} &= 3033.3333, \\
833.3333 + Q_{31}^{3(s)} &= H_{31}^{3(s)}, \\
H_{31}^{3(s)} + Q_{32}^{3(s)} &= H_{32}^{3(s)}, \\
H_{32}^{3(s)} + Q_{33}^{3(s)} &= H_{33}^{3(s)}, \\
H_{33}^{3(s)} + Q_{34}^{3(s)} &= 3000, \\
H_{jt}^{3(s)} &\leq 10000, \quad (j = 1, 2, 3; t = 1, 2, 3, 4).
\end{aligned}$$

6. Conclusion

This paper presented a methodology to solve FFMMLP problems based on the bound-decomposition method and harmonic mean technique. This methodology uses a crisp linear approach to convert FFMMLP problems into three crisp multilevel multiobjective linear programming (CMMLP) problems. Each crisp problem's objective functions can be solved independently, and their optimum values can be used to compute the fuzzy harmonic means FHM of these values. Thus, each CMMLP problem can be converted to a single objective. Then, a single objective linear programming problem is solved to find the final solution. Two examples are given to test the validity of this proposed method. One example is the proposed approach to a multilevel multiobjective production planning model for a supply chain under a fully fuzzy environment. The results obtained show the applicability and accuracy of the proposed technique. The proposed approach is excellent for solving FFMMLP problems because it can serve

decision-makers by providing an appropriate best solution to various multilevel multiobjective linear programming models with crisp or fuzzy parameters and variables simply and effectively. This procedure can be extended to solve fully rough multilevel multiobjective linear programming problems. This approach can also be extended to solve fully fuzzy rough multilevel multiobjective linear programming problems.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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